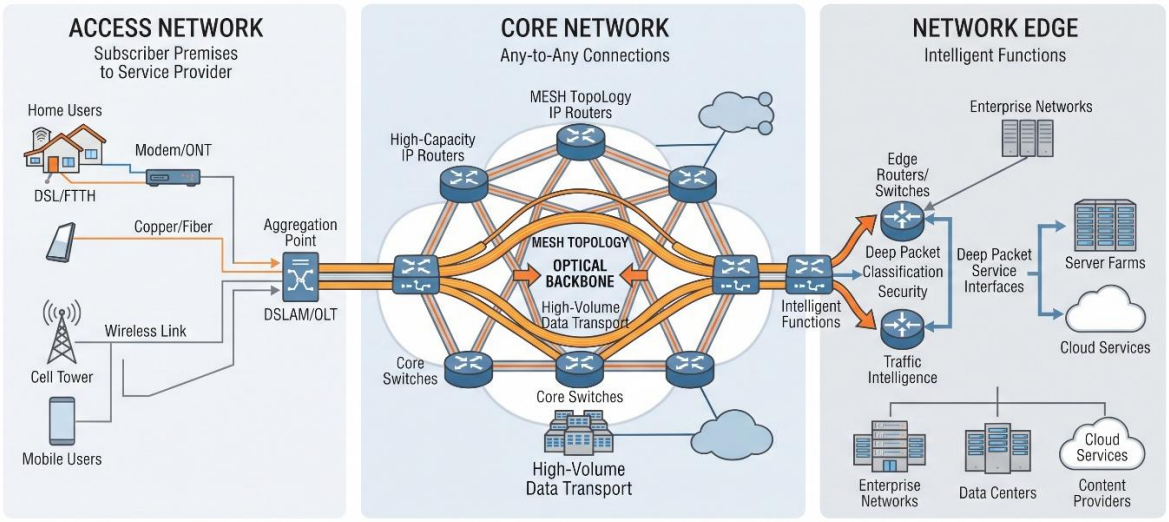
Network Infrastructures

Access Control

Modern communication networks are structured into distinct functional layers that work together to transport data from end-users to global destinations. Understanding this architecture requires examining three primary components: the **Access Network**, the **Core Network**, and the **Network Edge**.



Access Network

The access network is the foundational segment of a communications infrastructure that physically connects end-user devices to their immediate service provider (also known as the Internet Service Provider or local carrier).

Historically, telecommunication carriers relied primarily on traditional copper lines (twisted-pair telephone wires) designed for analog voice telephony. Today, these same physical lines—along with hybrid coaxial-fiber setups—are utilized to deliver high-speed **broadband internet** alongside traditional telephony signals. Due to the exponential growth in bandwidth demand, carriers are heavily investing in **optical fiber** deployments, which offer high data rates, immunity to electromagnetic interference, and stable performance over long distances.

Core Networks

The core network functions as the high-capacity backbone of the telecommunications infrastructure. It is typically structured in a mesh topology to guarantee robust, any-to-any connectivity (? A mesh layout provides multiple redundant paths between nodes. If one link or router fails, traffic can be dynamically rerouted through alternative paths, ensuring high availability and fault tolerance).

The core consists of high-performance IP routers and core switches interconnected by a high-speed optical backbone. By leveraging fiber optics at the core level, carriers can transport massive aggregates of data traffic across cities, countries, and continents with minimal latency and packet loss.

The Network Edge

The network edge refers to the boundary where end systems (hosts, servers, enterprise networks) interface with the core network. The edge may perform intelligent functions that are not performed inside the core network. In this scenario we divide:

- **Edge Devices (Switches/Routers):** These devices perform deep packet inspection or examine packet headers to classify traffic, apply security policies, shape traffic, and make initial routing decisions based on specific packet properties.

- **Core Devices:** Once the edge has processed and directed the traffic onto a chosen path, the core network focuses exclusively on high-speed, high-volume packet switching and forwarding without performing complex inspections.

This functional separation significantly enhances overall network performance. By keeping the core lightweight and focused entirely on rapid forwarding, bottlenecks are minimized, allowing the network to scale effectively to handle massive traffic volumes.

Types of Access

Wired: This category uses physical metallic or glass cables to establish a connection to the network. It guarantees high reliability, stable performance, and fast transmission speeds.

- **Powerline Access**: This technology uses existing electrical wiring within a building for network connectivity. It offers an easy setup without requiring new cables, making it useful for home networking and extending Wi-Fi coverage. However, it offers the least stable and lowest-performing connection quality among wired options.
- **DSL (Digital Subscriber Line) Access**: leverages traditional copper telephone lines to transmit digital data alongside voice signals. While constrained by distance from the telephone exchange, it offers the advantages of widespread availability and cost-effectiveness, making it historically popular for home internet and small businesses.
- **Cable Access**: Cable access utilizes coaxial cables—originally deployed for television broadcasting—to deliver bundled internet and TV services. It relies on a shared infrastructure among local subscribers to provide high-speed residential broadband.
- **Fiber-Optic Access**: Utilizing thin glass or plastic fibers to transmit data via light pulses, fiber-optic access delivers exceptional performance. Its primary advantages are high bandwidth, low latency, and high security, making it the standard for high-speed internet backbones and data centers.

Wireless: Wireless technologies use radio signals to connect devices to the network. Their key advantages include user mobility, architectural flexibility, and quick deployment. Typical examples range from local Wi-Fi networks to short-range Bluetooth connections.

- **Satellite Access**: connects remote terminals to communication satellites orbiting the Earth. Although it provides unmatched geographic coverage for remote and rural areas, traditional satellite links suffer from inherently high propagation latency (due to the vast distance to space) and environmental susceptibility.
- **Cellular Access**: Connecting via mobile network base stations, cellular access ensures continuous mobility, flexibility, and on-the-go connectivity. It is predominantly utilized by smartphones and mobile devices for wide-area cellular data, but struggle with bandwidth limitations and variable signal strength depending on user location and cell tower congestion.
- **5G Access**: Representing the fifth generation of mobile networks, 5G offers dramatically enhanced performance with ultra-high speeds and ultra-low latency. Beyond human-centric communications, it supports massive machine-type communications for the Internet of Things (IoT) and immersive technologies like Augmented/Virtual Reality (AR/VR), making it foundational for smart cities.

Wired Access

Copper Access

The copper access network was historically developed to support voice communications, but then evolved to meet both traditional telephone services and high-speed broadband.

To achieve this, operators implement digital subscriber line standards such as ADSL (Asymmetric Digital Subscriber Line) and VDSL (Very-high-bit-rate Digital Subscriber Line), which separate the transmission medium into discrete frequency bands. In addition, the major advantage of this network is its widespread availability and the use of existing infrastructure.

However, existing copper-based networks have gradually been expanded during several decades and their architectures are not optimized with regard to use of current technologies. If an entirely new network were to be built today, it would not be based on use of copper-based technologies (in favor of optical media), and the design would therefore be very different from those of today's copper-based networks.

One problem is that networks are designed mainly for carrying POTS, while a growing share of the traffic is based on IP or other data communication protocols, and in some areas, there are problems with capacity and quality of service.

Optical Access

Optical access systems deliver ultra-high speeds with high reliability, making them the primary choice for fixed access networks. Unlike copper, optical transmission eliminates distance-related attenuation and electromagnetic interference. The main technology types used depend on the deployment scenario:

- **GE-PON (Gigabit Ethernet Passive Optical Network):** mainly adopted for **individual** residential users.
- **G-PON (Gigabit Passive Optical Network):** typically deployed for **enterprise** and business customers.
- **CDM (Code Division Multiplexing over Fiber):** used to cover **long-distance** transmissions.

FTTx (Fiber-to-the-x)

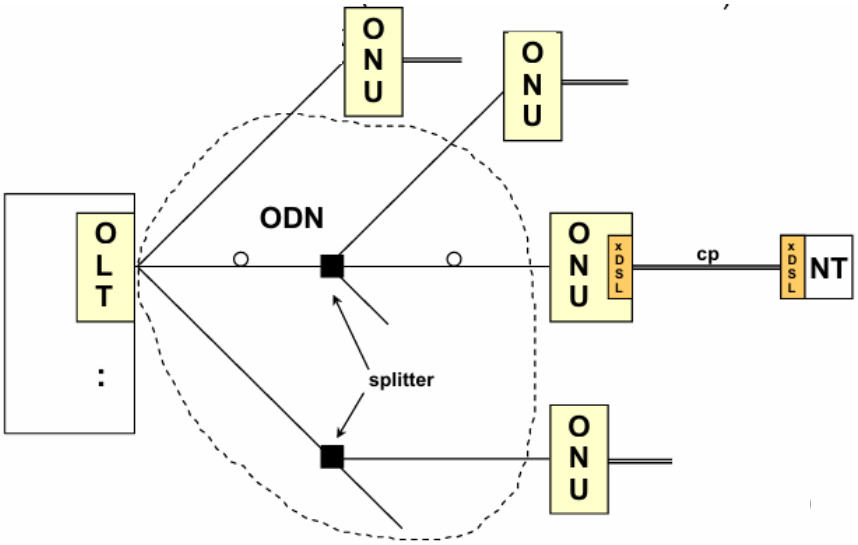
While technologies like PON define *how* data is transmitted, FTTx (Fiber-to-the-x) describes the physical deployment architecture—specifically, how close the optical fiber gets to the end user before switching to legacy copper wiring or terminating directly at the subscriber. These architectures replace traditional copper infrastructure by running fiber from a network element, such as the central office or local exchange, directly to a target destination denoted by x:

- **FTTH (Home):** The fiber runs all the way into the individual residence.
- **FTTP (Premise) / FTTU (User):** The fiber extends directly to the specific property or user.
- **FTTB (Building / Business):** The fiber reaches the boundary of the building (usually a basement or telecom room), while internal cabling handles distribution.
- **FTTO (Office) / FTTD (Desk):** The fiber reaches deep inside commercial environments, down to individual office rooms or desks.
- **FTTC (Curb):** The fiber reaches a street cabinet near the pavement/curb, with existing copper covering the final short distance.
- **FTTN (Node / Neighborhood) / FTTZ (Zone):** The fiber terminates at a shared distribution node serving a larger neighborhood or zone.

FTTx Elements

An FTTx deployment relies on four standard building blocks to transmit data between the service provider and the customer:

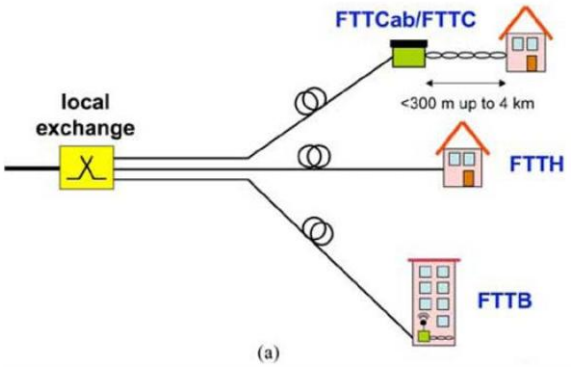
- **OLT (Optical Line Terminal):** The endpoint located at the service provider's Central Office (CO). It controls the bidirectional flow of information across the optical network and connects to the core IP network.
- **ODN (Optical Distribution Network):** The physical transmission path connecting the OLT to the ONUs, consisting of fiber cables, connectors, and splitters.
- **ONU (Optical Network Unit):** The device placed near or at customer premises that converts optical signals into electrical signals for end-user equipment.
- **ONT (Optical Network Termination):** A specific type of ONU deployed directly at the customer premises to serve as the local network termination point.



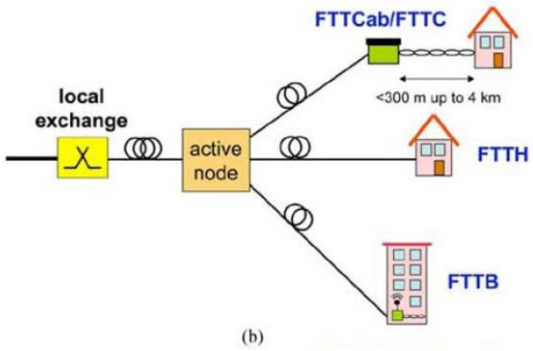
FTTx Architectures

FTTx networks are deployed using three main architectural models based on how the transmission medium is shared and managed:

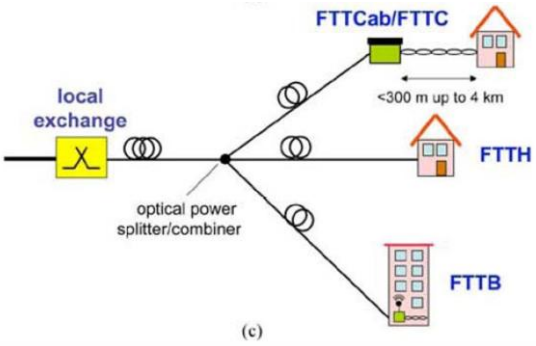
- **AON (Active Optical Network):** Also known as Point-to-Point (P2P), this setup provides a dedicated optical fiber directly from the central office switch to each individual subscriber, ensuring the highest network performance. However, this method costs too much to be really used in a normal scenario. As we can see in the picture (a) it requires a specific dig for each user (3 users => 3 digs) increasing the cost. So, this method costs so much due to the high cost of digging, (about 70 euros per meter, while the actual cost of fiber is extremely low, around 0.5 cents per meter.



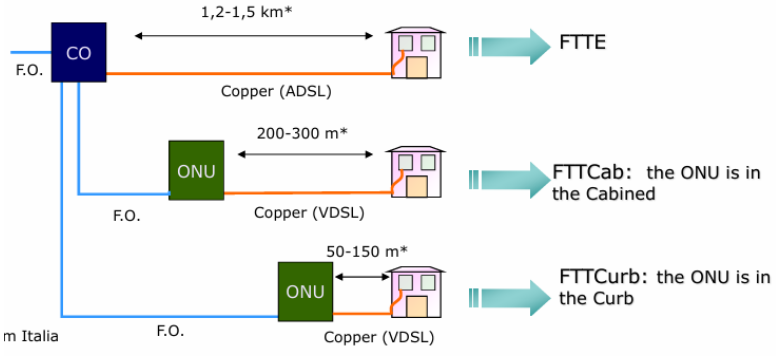
An intermediate solution (b) relies on electrically powered switching (the active node) that receives the light from the fiber, convert it into electricity, operates as a "router" and then retransmits light to the users. However, this method removes the advantage of dedicated fiber lines (the line from the local exchange to the active node it's shared with the neighbourhood) and introduces operational drawbacks: the active nodes require electrical power, hardware maintenance, dedicated physical space, and represent potential single points of failure. Furthermore, dedicated runs increase deployment costs significantly (e.g., 3 endpoints require 3 separate fibers and potentially 3 trenching digs).



- **PON (Passive Optical Network):** It's the most used architecture. It's a point-to-multipoint architecture based on a **tree topology**. Instead of dedicated fiber or powered intermediate switches, it uses unpowered **optical splitters** to passively divide a single optical signal from the OLT among multiple end users. This drastically reduces cabling requirements and maintenance costs since it doesn't need electricy or space (no more active node, no switching, routing, no light->electricity->light, etc…). Practically, the splitter is a simple piece of fiber (usually made of glass) fused with the other fiber cables. In terms of performances modern PON standards deliver high speeds (from 1 Gbps up to 10 Gbps shared), easily meeting typical residential and business needs. However, individual performance can fluctuate slightly during peak traffic hours because users share the aggregate optical channel.



- **Fiber to the Exchange:** A hybrid legacy model where optical fiber extends only as far as the Central Office (CO). The connection from the Central Office to the end customer is completed entirely over existing copper lines using digital subscriber technologies like ADSL. The primary disadvantage here is the copper local loop itself, which severely limits bandwidth, increases attenuation over distance, and suffers from electromagnetic interference. It's the worst in term of performances: Signal attenuation increases sharply over distance, drastically reducing speeds as the subscriber gets farther from the exchange. Speeds are low and highly asymmetrical compared to all-optical solutions.



Wireless Access

The wireless access domain represents the sector holding the greatest promise for realizing the concept of ubiquitous networking. Its primary objective is to ensure communications without interruptions (*seamless*) and high-speed broadband connectivity by integrating both fixed and mobile access into a single architecture.

Talking about wireless access solution we must distinguish between:

- **Fixed Wireless Access**: instead of having cables interconnecting users and providers, FWA uses antennas. So, we have an antenna on the customer's building (e.g., a house or school) that communicates directly with the central provider antenna, located at a fixed base site.
- **Cellular**: A more complex infrastructure designed to provide connectivity to moving users over large geographic areas by distributing the coverage across adjacent radio cells.

Cellular Networks

Cellular networks are designed to provide wireless connectivity across **broad geographical regions** while actively managing **user mobility** and dedicated radio resources.

Even if these systems are named "Wireless Access" **the wireless segment is actually very small**:

- Only the radio hop between the end user and the base station is wireless.
- The rest of the infrastructure, connecting the base station antennas back to the core network, also known as the **backhaul** is completely wired.

Historically, the backhaul relied on copper lines (E1/T1 links up to 2G/3G), but it has since evolved to high-speed optical fiber to support growing data rates.

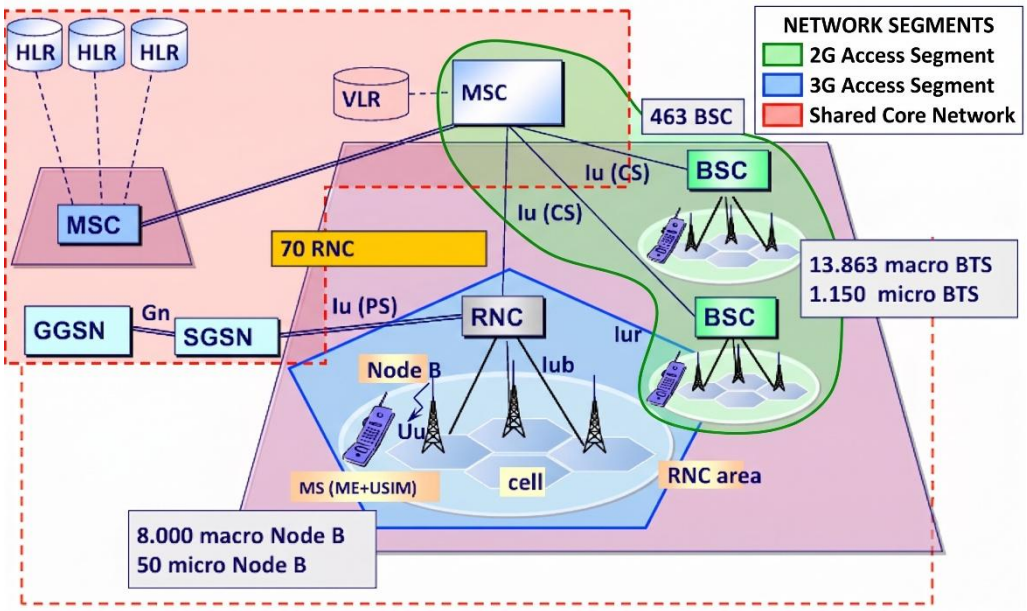
Characteristics

Cellular networks rely on two critical network operations:

1. **Dedicated Bandwidth Management:** Unlike Wi-Fi, which operates on a shared contention-based, in a Cellular Network the transmission resources (time, frequency, or codes) are allocated individually to each active user, ensuring strict quality of service and throughput management.
2. **Mobility Management:** Because cellular users can move freely (e.g., walking or traveling on a train), the network must support continuous connectivity through:
    - **Localization:** The network continuously tracks where the user terminal is located (Wi-Fi only associates a client with a local Access Point and does not know its wider location).
    - **Handover:** The connection must actively "follow" the user switching active data sessions from one antenna to another without dropping the service.

2G and 3G Architecture

In early cellular generations (2G and 3G, as well as 4G), the network divides the territory into geographical areas called **cells** (the name "Cellular" derives exactly from this. Cells => Cellular). Each cell is served by a Base Station (the physical antenna site.



- **2G Access Segment**: The mobile terminal communicates via radio with a base station known as a BTS (Base Transceiver Station). Multiple BTS units are aggregated and controlled by a hierarchical controller called a BSC (Base Station Controller).
- **3G Access Segment**: The user device (referred to as an MS - Mobile Station) connects via the Uu radio interface to the 3G base station, known as a Node B. Node Bs are managed and coordinated by an RNC (Radio Network Controller) via the Iub interface. Additionally, multiple RNCs can communicate directly with one another through the Iur interface to manage mobility and handovers.
- **Shared Core Network**: Both generations connect to a core network divided into two functional domains:
    - **Circuit-Switched (CS) Domain**: Dedicated primarily to traditional voice traffic, it centers on the MSC (Mobile Switching Center), supported by the HLR (Home Location Register) and VLR (Visitor Location Register) databases for location and user profiles. Both the BSC (2G) and the RNC (3G) interface with the MSC via dedicated links (the Iu(CS) interface for 3G).
    - **Packet-Switched (PS) Domain**: Dedicated to data traffic and Internet access; introduced with GPRS/EDGE and consolidated in 3G. The RNC connects via the Iu(PS) interface to the SGSN (Serving GPRS Support Node), which communicates via the internal Gn interface with the GGSN (Gateway GPRS Support Node) to route packets to external IP networks.

5G Architecture

In fifth-generation (5G) networks, the architecture is significantly simplified and modernized into a **Service-Based Architecture (SBA)** which clearly separates the control plane from the data transport plane. Like the older generations, the only wireless link remains the connection between the end user and the antenna, while the internal network interfaces (e.g., N2, N3, N4, N6) are high-speed wired links (usually fiber).

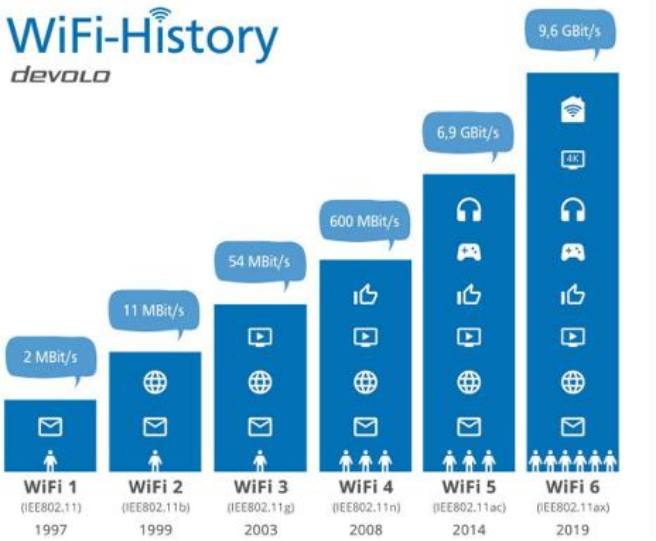


The key components include:

- **UE (User Equipment):** The end-user device (such as a smartphone, laptop, or IoT sensor) requesting network connectivity.
- **gNB (next-generation Node B):** The 5G base station antenna that manages the radio interface directly with the UE.
- **AMF (Access and Mobility Management Function):** responsible for authenticating the UE, handling network registration, and managing user mobility between gNB antennas.
- **SMF (Session Management Function):** A control entity that works alongside the AMF to allocate resources to each session.
- **UPF (User Plane Function):** It processes and routes data packets between the 5G radio access network and the external data network (Internet/Data Network). It's the part in which the packets are switched through the network to reach the end user.

Wi-Fi Evolution

The IEEE 802.11 family has progressively evolved over the past decades to address rising bandwidth demands, higher client densities, and media-rich services. To simplify technical nomenclature, the Wi-Fi Alliance retrospectively introduced user-friendly generation names (Wi-Fi 1 through Wi-Fi 6).



- **Wi-Fi 1 (IEEE 802.11 – 1997):** The legacy baseline standard offering a peak throughput of **2 Mbit/s**. It was primarily suited for basic, low-bandwidth text transmission like email.
- **Wi-Fi 2 (IEEE 802.11b – 1999):** Raised the maximum data rate to **11 Mbit/s** over the 2.4 GHz band. This standard enabled the early expansion of wireless home networking and general web browsing.
- **Wi-Fi 3 (IEEE 802.11g – 2003):** Introduced OFDM (Orthogonal Frequency-Division Multiplexing) in the 2.4 GHz band, increasing peak theoretical speed to **54 Mbit/s**. This allowed users to support standard video streaming along with browsing and email.
- **Wi-Fi 4 (IEEE 802.11n – 2008):** A major evolutionary step reaching up to **600 Mbit/s**. It introduced MIMO (Multiple-Input Multiple-Output) antenna technology and dual-band operation (2.4 GHz and 5 GHz), enabling multi-user environments, social media use, and higher-resolution video streaming.
- **Wi-Fi 5 (IEEE 802.11ac – 2014):** Focused heavily on the 5 GHz band, leveraging wider channels and MU-MIMO to push maximum throughput up to **6.9 Gbit/s**. This bandwidth headroom unlocked reliable online gaming, high-fidelity audio, and dense multi-device connectivity.
- **Wi-Fi 6 (IEEE 802.11ax – 2019):** Designed to deliver up to **9.6 Gbit/s**. Beyond raw speed, its main improvement is spectral efficiency in crowded environments using OFDMA (Orthogonal Frequency-Division Multiple Access), making it capable of supporting 4K/8K media streams and high client densities simultaneously. It also allows us to make calls on the mobile network (e.g., if you are abroad and have no roaming active).

GEO Satellites Access

GEO (**Geostationary Orbit**) is a type of access that relies on satellites which orbit at an altitude of approximately **35,786 km** directly above the Earth's equator. At this specific distance, their orbital speed matches the rotation of the Earth, which means a GEO satellite appears completely stationary in the sky from the perspective of a ground observer.

Because they are positioned so high, just **three satellites** are mathematically sufficient to provide near-global coverage and ground antennas do not need to track them across the sky (since the satellites seem stationary from Earth, the antennas simply “point at a fixed location in the sky”).

However, this massive distance introduces an inevitable physical delay with a transmission latency of **500 ms or more**, making GEO unusable for today’s real-time interactive traffic (like voice calls, online gaming, or rapid two-way communication).

LEO Satellites Access

LEO (**Low Earth Orbit**) solves the latency problems of GEO by relying on much closer satellites, which orbit at an altitude of approximately **500 to 2000 km** above Earth (15-20 times closer than GEO).

However, due to their proximity to the surface, they have a short orbital period of roughly **90 to 120 minutes** per orbit, meaning a single satellite cannot provide persistent local service. To solve this problem and guarantee continuous global coverage LEO requires **large satellite constellations** and specific algorithms that move clients from a satellite to another without connection loss.

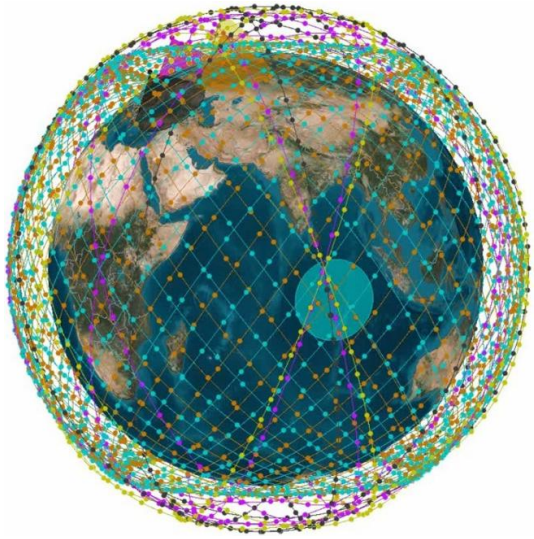
- **Advantages:**
    - **Low Latency:** Round-trip transmission time is roughly **20–40 ms**, which is comparable to terrestrial broadband and drastically lower than legacy satellite links.
    - **High Bandwidth Potential:** Proximity enables strong signal reception and higher throughput capacity.
    - **Global Reach:** Delivers reliable coverage to remote, rural, and maritime areas underserved by fiber or mobile networks.

- **Challenges:**
    - **Constellation Size:** Thousands of active satellites are required to ensure continuous handovers without dropped connections.
    - **High Deployment and Maintenance Costs:** Launching, managing, and replenishing a massive constellation involves major capital investment.
    - **Short Lifespan and Space Debris:** Operating in low orbit subjects spacecraft to atmospheric drag, resulting in a short operational lifespan of only **5 to 7 years** and increasing orbital debris risks.

LEO vs. GEO

Feature	LEO (Low Earth Orbit)	GEO (Geostationary Orbit)
Altitude	~500–2,000 km	~35,786 km
Latency	Low (~20–40 ms)	High (~500+ ms)
Coverage Mechanism	Requires thousands of satellites in a constellation	A single satellite covers roughly 1/3 of the Earth
Lifespan	Shorter (~5–7 years)	Long (~15 years)

Example: Starlink



Developed by SpaceX, **Starlink** represents the most prominent commercial deployment of a broadband LEO mega-constellation:

- **Scale and Purpose:** Over 5,000 satellites deployed to deliver global broadband coverage to residential users, enterprises, and mobility sectors (such as aircraft and maritime vessels).
- **Terminal and Performance:** End users connect via a compact, phased-array antenna terminal ("Dishy"), achieving typical round-trip latencies of **20–40 ms**.

Non so se va fatta questa parte. La Cuomo ha detto che lo vedremo meglio con l'altro prof (Polverini). Quindi questa potrebbe essere una semplice introduzione che rivedremo poi.

Backbone Networks

A telecommunications backbone is split into two distinct architectural views:

- **Logical Topology**: which describes how IP traffic and routing protocols see the network).
- **Physical Topology**: the underlying level that actually contains fiber optics and transmission equipment in the ground.

Logical Topology: The Optical Packet Backbone (OPB)

The Optical Packet Backbone (OPB) represents the packet-switched routing layer consisting of 32 Points of Presence (PoPs) distributed across the country. It is organized in a hierarchical, two-tier star/mesh structure to aggregate and forward IP traffic efficiently:

- Inner Core (4 PoPs): The central transit backbone consists of 4 high-capacity PoPs: 2 located in Rome and 2 in Milan. These main nodes are interconnected with heavy trunk links operating at 10 Gbit/s (STM-64).
- Outer Core (28 PoPs): The remaining 28 regional PoPs aggregate local traffic and connect back to the Inner Core. Depending on local traffic demand, these links utilize different transmission rates:
    - 10 Gbit/s (STM-64) for high-volume secondary nodes.
    - 2.5 Gbit/s (STM-16) for intermediate regional nodes.
    - 155 Mbit/s (STM-1) for lower-demand peripheral links.
- Dimensioning Rule (50% Link Utilization): Core links are intentionally engineered to operate at an average load of 50% capacity. This ensures adequate headroom to absorb traffic spikes and allows immediate failover rerouting without congestion if an alternate link goes down.

Physical Topology: The Optical Transport Network (OTN)

The logical OPB does not run on dedicated end-to-end dark fiber for every single logical link; instead, it is built directly on top of an underlying physical Optical Transport Network (OTN).

- **Physical Nodes (ODXC):** Physical transmission sites house ODXC (Optical Digital Cross-Connect) equipment. These devices switch high-capacity lightpaths and groom optical wavelengths directly at the physical layer without needing packet inspection.
- **Multiplexed Transmission (DWDM):** Long-distance physical spans use DWDM (Dense Wavelength Division Multiplexing) over optical fiber. DWDM allows many individual optical channels (lambdas) to travel down a single physical pair of fibers simultaneously.
- **Logical to Physical Independence:** The logical map looks like a direct, simplified tree/star radiating from Rome and Milan. However, the physical map reveals a heavily meshed ring topology connecting physical cities (e.g., Turin, Genoa, Florence, Bologna, Naples, Palermo). If a physical optical link is severed, the DWDM/ODXC layer can automatically reroute the optical channel through an alternate geographic path while the logical IP layer above remains completely unaware of the physical fault.

DSL

Def. DSL represents a family of technologies designed to provide **digital data transmission over the wires of a local telephone network** (twisted-pair copper wires).

Historically, the acronym stood for **Digital Subscriber Loop**, but it was later rebranded as **Digital Subscriber Line** for marketing purposes.

(A "subscriber line" refers to the physical telephone connection that links an end customer, the subscriber holding a service contract with a network operator, directly to the operator's central office).

The xDSL Family

Over time, several variants of DSL were developed to meet different performance requirements and line conditions:

- **IDSL (ISDN Digital Subscriber Line):** Deployed to provide digital access on very old telephone lines.
- **HDSL (High Data Rate Digital Subscriber Line):** An early high-speed variant offering symmetrical transmission.
- **SDSL (Symmetric Digital Subscriber Line):** A standardized, single-pair evolution of HDSL delivering identical upstream and downstream speeds.
- **ADSL (Asymmetric Digital Subscriber Line):** Configured with higher download speeds and lower upload speeds to match typical consumer internet habits; it became the dominant commercial standard of its era.
- **ADSL "lite" (or G.lite):** A simplified, lower-speed version of ADSL that eliminated the need for a splitter installed at the subscriber's premises.
- **RADSL (Rate-Adaptive Digital Subscriber Line):** A dynamic variant that automatically adjusts its transmission rate based on the physical quality and length of the copper line.
- **VDSL (Very High Speed Digital Subscriber Line):** Designed to deliver substantially higher bitrates over shorter physical distances.
- **VDSL2 (Very High Speed Digital Subscriber Line 2):** An improved, wider-bandwidth extension of VDSL that became widely adopted alongside fiber-to-the-cabinet (FTTC) deployments.

Motivation

Globally, telecommunications infrastructure includes more than 600 million copper wire connections, spanning hundreds of millions of kilometers. This vast network was initially built strictly for analog voice communication, in which a continuous electrical waveform represents acoustic sound.

The transition from analog voice to digital data transmission, encoding information as bits rather than electricity, was driven by several market and technical factors:

- **Surging Data Demand:** The rapid expansion of personal computing created a massive requirement for high-speed access to packet-switched IP networks, that analog systems could not provide.
- **Competition and Deregulation:** The liberalization of national telecommunication markets allowed third-party operators to compete with historical incumbents, driving the commercial rollout of broadband services.
- **Global Telephony Demands:** Ongoing requirements to expand basic POTS (Plain Old Telephone Service) connectivity in developing regions, where physical copper lines remained the primary telecommunications asset.

Analog Dial-up Modems

Before DSL, transmitting digital data over the telephone network required an **analog modem**.

- **Dial-up Mechanism:** The user's computer connected to a local modem, which converted digital data into audible analog audio tones. The signal traveled over the UTP line through the Central Office switch and traversed the circuit-switched PSTN to reach another modem and router operated by the ISP (Internet Service Provider).
- **Line Occupancy:** Because it utilized the standard voice channel, a dial-up connection occupied the line completely, preventing simultaneous voice calls.

Physical Limits: Shannon Channel Capacity

The maximum achievable bit rate of an ordinary analog telephone line is strictly bounded by the Shannon-Hartley theorem:

$$C = B \log_2(1 + \text{SNR})$$

Example: In a standard analog voice circuit, low-pass filters inside the telephone network restrict the available bandwidth to narrow voice frequencies:

- **Bandwidth (B or W):** Approximately 3.4 kHz (3400 Hz).
- **Signal-to-Noise Ratio (**SNR**):** Typically around $10,000$, which corresponds to:

$$10 \log_{10}(10000) = 40 \text{ dB}$$

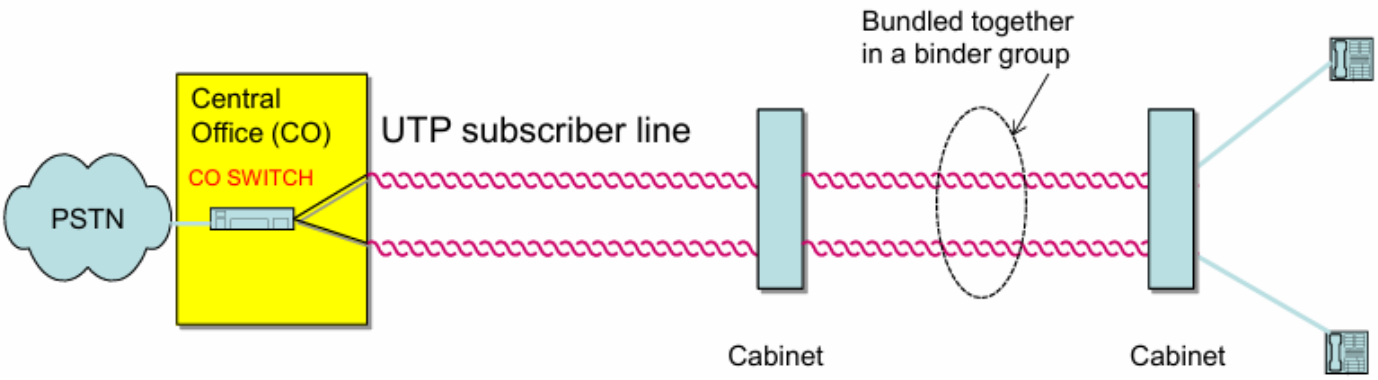
Applying the formula:

$$C = 3400 \cdot \log_2(1 + 10000) \approx 44,800 \text{ bits/second}$$

So, 44.8 kbps is the upper theoretical capacity of a traditional analog voice channel…

It explains why dial-up analog modems could not exceed speeds of roughly 33.6 kbps to 56 kbps, as long as the data was forced to pass through the voice-band filters of the PSTN. And this is also why we should move beyond this technology.

Pre-DSL Wiring Schematic



Before the introduction of digital subscriber line (DSL) technologies, the local loop was deployed exclusively to connect end-user telephones to the circuit-switched telephone network:

- **UTP Subscriber Line:** The connection from the customer premises to the Central Office (CO) consists of an unshielded twisted-pair (UTP) subscriber line.

- **Cabinets and Binder Groups:** Along the transmission route, subscriber lines pass through intermediate street cabinets where individual copper pairs are bundled together into large binder groups. Because multiple pairs are packed closely within the same cable binder, electromagnetic interference (crosstalk) becomes a significant physical constraint.
- **Core Connection:** At the Central Office, all incoming subscriber lines terminate at the **CO switch**, which directly routes calls into the legacy **PSTN** (Public Switched Telephone Network).

Asymmetric DSL

L'ADSL (Asymmetric Digital Subscriber Line) è stata concepita originariamente con l'obiettivo di distribuire contenuti di video on demand (VoD), anche se questo target iniziale non ha avuto successo commerciale. Il punto di forza dell'ADSL risiede nel fatto che il doppino in rame (il local loop) è a **uso esclusivo del singolo abbonato**, senza alcuna contesa di banda sul segmento di accesso fisico. Inoltre, la tecnologia garantisce la trasmissione passiva del servizio telefonico analogico tradizionale (**POTS**), consentendo l'uso contemporaneo di voce e dati sulla stessa linea.

La caratteristica distintiva dell'ADSL rispetto alle altre varianti DSL è l'asimmetria del flusso di dati:

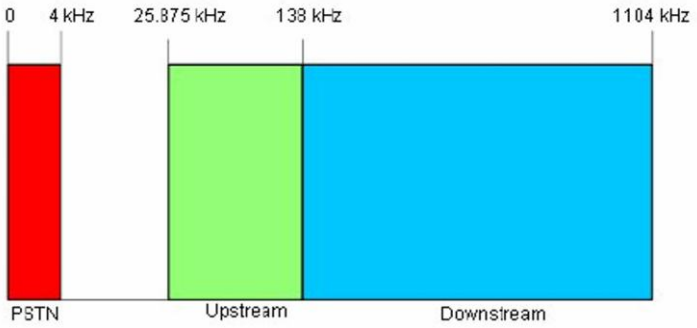
> Il volume e la velocità di trasmissione sono nettamente maggiori in una direzione rispetto all'altra.

Questo soddisfa a pieno le esigenze dell'utente medio: **throughput elevato in download** (navigazione sul web, streaming video, download di pagine HTML, etc…) ma **capacità ridotta in upload** (non dovendo ospitare server o trasmettere flussi continui verso la rete).

ADSL Frequency Bands

A frequenze elevate il rumore e l'attenuazione aumentano drasticamente. Per gestire in modo efficiente la trasmissione bidirezionale senza che i flussi collidano sullo stesso mezzo, lo spettro viene suddiviso in canali a frequenze separate (**FDD - Frequency Division Duplexing**):

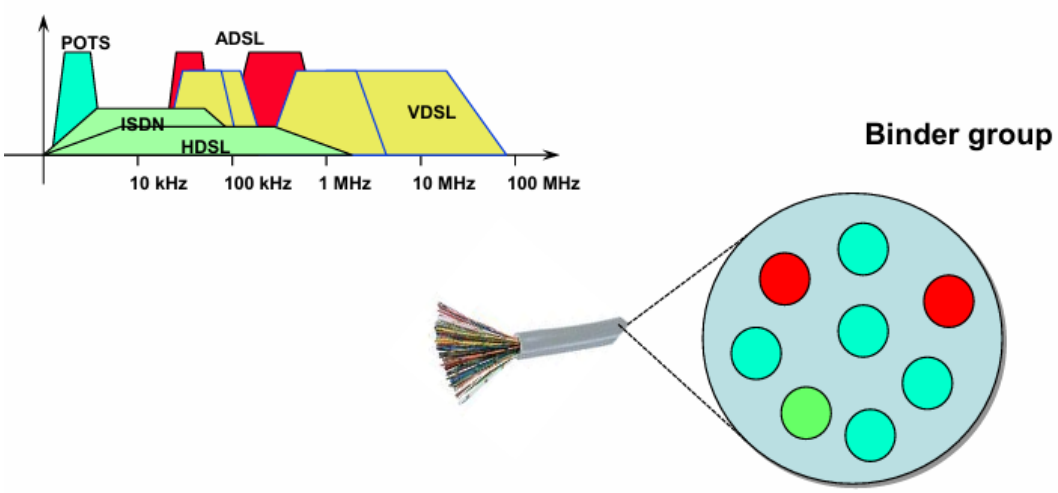
1. **Canale Fonico (POTS):** Mantiene la banda base analogica (0 – 4 kHz) per le chiamate vocali.
2. **Upstream Band:** Copre la banda da **25.875 kHz a 138 kHz**, riservata alla trasmissione dall'utente verso la centrale telefonica (Central Office).
3. **Downstream Band:** Parte da frequenze superiori a 138 kHz e si estende verso l'alto; occupando un intervallo spettrale molto più ampio rispetto all'upstream, garantisce una velocità di download nettamente superiore.



Cross-talks

Uno dei problemi principali dell'ADSL riguarda le Cross-Talks causate dai doppini in rame.

Infatti, i doppini di rame non viaggiano isolati, ma sono solitamente raggruppati in "**binder groups**" posati nelle stesse canaline sotterranee. Tuttavia, quando più coppie fisiche condividono lo stesso cavo e utilizzano bande di frequenza sovrapposte, il campo elettromagnetico generato da un segnale si accavalla con gli altri doppini di rame, generando cross-talk (interferenza).



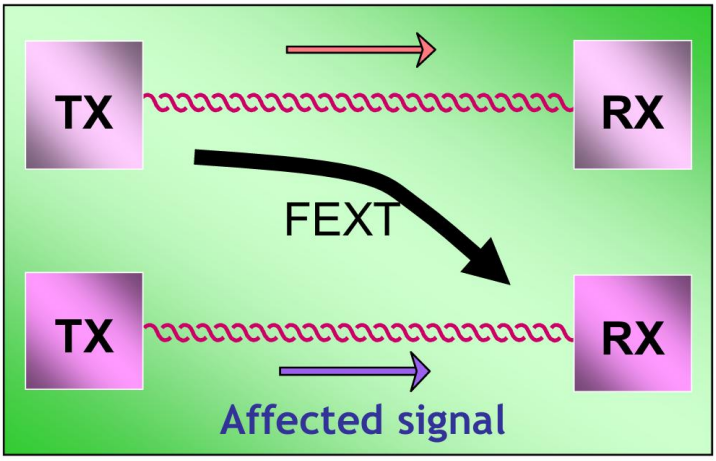
Il crosstalk cresce significativamente all'aumentare della frequenza e dipende da tre fattori principali:

- La densità spettrale di potenza (PSD) del segnale trasmesso.
- Il numero di coppie ritorte presenti all'interno del medesimo cavo.
- Il grado di sovrapposizione tra le bande di frequenza del segnale utile e di quelli interferenti.

FEXT (Far-End Cross-Talk)

FEXT è un tipo di Cross Talk che si verifica tra un trasmettitore e un ricevitore posti su estremità opposte del cavo. Tipicamente il segnale interferente percorre l'intera lunghezza del canale perdendo potenza lungo il tragitto, prima di causare interferenze vere.

Pertanto, il FETX tipicamente non è un problema nelle reti abbastanza lunghe, m apuo essere fastidioso su reti ADSL relativamente brevi. Per limitare questo fenomeno, la dimensione dei fasci viene regolata limitando il numero di coppie per condotto (di solito non più di 12 doppini di rame).

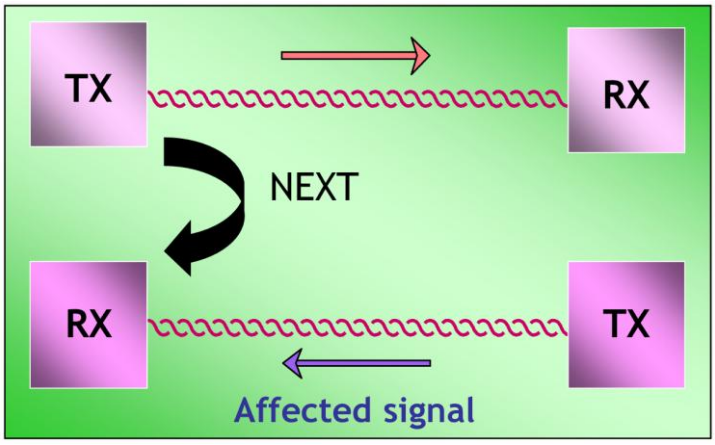


NEXT (Near-End Cross-Talk)

NEXT è un tipo di Cross Talk che si verifica tra un trasmettitore e un ricevitore posti sullo stesso lato del cavo.

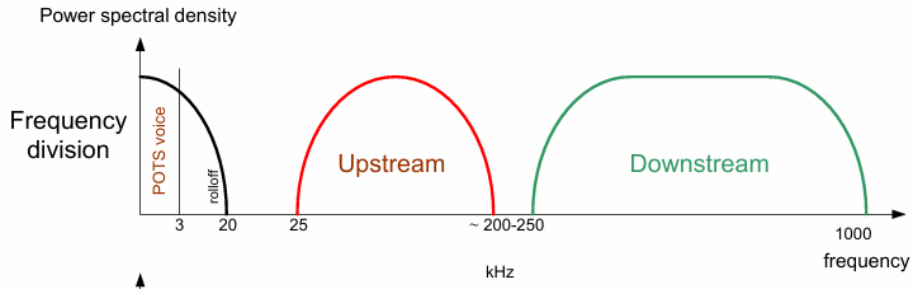
L'idea è che il segnale ricevuto dal ricevitore, provenendo da lontano, giunge al ricevitore con un livello di potenza già molto basso (a causa del lungo percorso). Al contrario, il trasmettitore, che si trova sullo stesso lato del ricevitore, trasmette un segnale alla massima potenza, coprendo il già debole segnale del ricevente. Per questo motivo, **il NEXT è molto più distruttivo del FEXT**.

NEXT is one of the reason of the frequency division for upstream and downstream in ADSL.

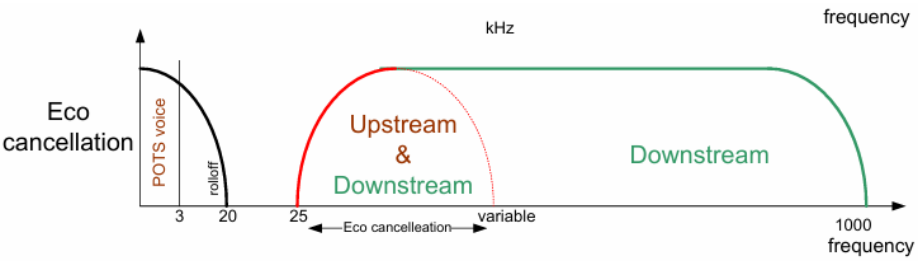


La separazione rigida in frequenza tra upstream e downstream nelle implementazioni ADSL standard è progettata proprio per **eliminare il NEXT**:

- Se tutti i trasmettitori posti allo stesso terminale (ad esempio tutti i modem degli utenti o tutte le porte del DSLAM in centrale) trasmettono solo su determinate bande e ascoltano solo su bande differenti, il NEXT non può manifestarsi perché il ricevitore filtra via le frequenze usate dai trasmettitori locali.



- **Caso critico (Sovrapposizione di banda):** Se si adotta una configurazione in cui la banda di downstream viene estesa verso il basso per aumentare ulteriormente il bitrate, andandosi a sovrapporre allo spettro di upstream dei clienti vicini nello stesso binder group, il NEXT torna a manifestarsi: il forte segnale downstream uscente dalla centrale o dai modem interferisce direttamente con i ricevitori adiacenti posizionati sulla medesima terminazione.



ADSL Filtering

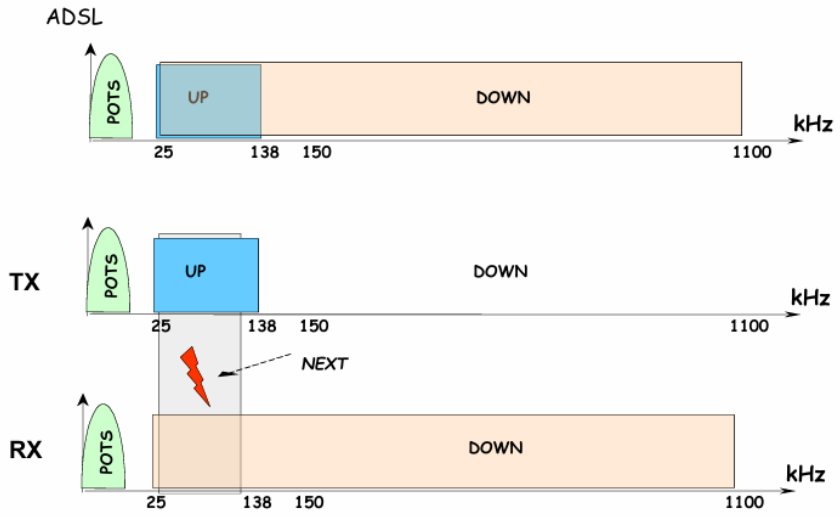
Because POTS and ADSL share the same physical copper pair across different frequency bands, appropriate frequency filters must be installed at the customer's premises to prevent high-frequency DSL signals from causing audible noise and interference on the telephone equipment:

- **Central Splitter**: A single device installed directly where the telephone line enters the building. It splits the incoming line into two dedicated paths before connecting to hardware: a low-pass path for the telephone and a high-pass path for the DSL modem.
- **Distributed Microfilters**: If a central splitter is not installed, an individual microfilter must be plugged into every telephone wall jack that has a phone connected. The microfilter blocks the high-frequency signals from entering the analog phone.

Echo Cancellation

In standard ADSL upstream and downstream use completely separated frequency bands to avoid interference. However, some ADSL configurations attempt to maximize downstream transmission rates by extending the downstream spectrum over the upstream band, accepting the risk that it could cause overlaps.

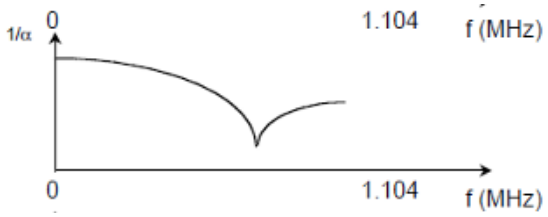
- **NEXT Problems:** as we've already seen, when upstream transmission and downstream reception share the same frequency band on nearby lines within the same cable binder, NEXT occurs. The strong local upstream transmit signal couples directly into the sensitive receiver circuit listening for incoming downstream traffic.
- **Local Echo:** In addition to interfering with neighboring pairs via NEXT, transmitting and receiving in the same frequency band causes the modem's own transmit signal to bleed into its receiver.
    - **Solution: Echo Cancellation.** To operate under this overlapping band configuration, modemes must implement Echo Cancellation. Echo cancellation uses digital signal processing to subtract a the transmitted signal from the combined incoming signal, allowing the receiver to recover the weaker incoming downstream data even in the presence of spectral overlap.

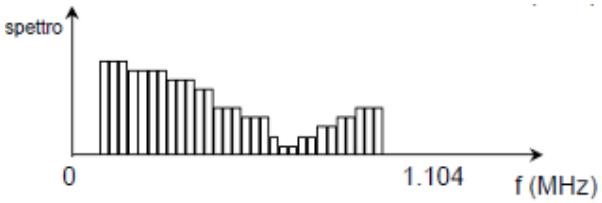


ADSL Modulations

Ideally, we would always like to have a flat frequency response because it guarantees uniform transmission characteristics across the entire spectrum, making signal detection and equalization simple.

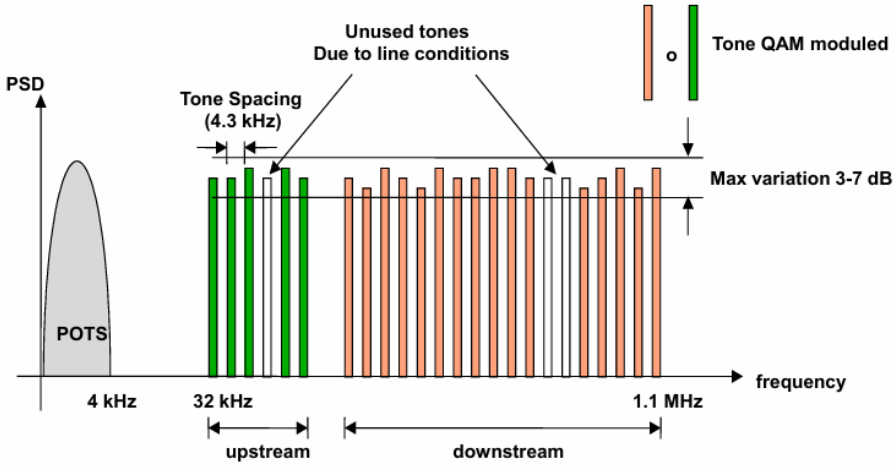
However, in copper lines signal attenuation and noise vary sharply across frequencies. So, when the bandwidth is large (such as in the ADSL case) the channel is not flat, causing severe signal distortion and complex equalization requirements.





To address this issue **Discrete Multi-Tone Modulation (DMT)** was introduced. This technique divides the total operational bandwidth into multiple independent sub-channels:

- **Flat Response Approximation:** Because each sub-band is extremely narrow, the channel behaviour in each sub-band it's almost flat, simplifying equalization and detection.
- **Per-Tone Modulation:** Each tone is centered around a discrete carrier and independently modulated. If a sub-channel experiences severe external interference, the system can disable it entirely (like the white ones in the image).
- **Dynamic Adaptation:** DMT continuously monitors line conditions and dynamically shifts bit allocation across sub-channels to maximize throughput without dropping the physical connection.



ADSL Theoretical Capacity

However, we must calculate the **maximum theoretical capacity** for each sub-channel using the Shannon capacity formula (studied some paragrapher ago):

$$C_t = B_t * \log_2(1 + SNR_t)$$

Where:

- C_t = Bit Rate of the tone *t*
- B_t = Bandwidth of the tone *t*
- SNR_t = Signal-Noise Ration of the tone *t*

In DMT, even if each tone has the exact same bandwidth $\boldsymbol{B_t}$, the $SNR_{\boldsymbol{t}}$ varies substantially across different frequencies, resulting in different bitrates for each tone.

However, assuming a theoretical maximum spectral efficiency of $\mathbf{15}$ bit/s/Hz per sub-channel, we obtain:

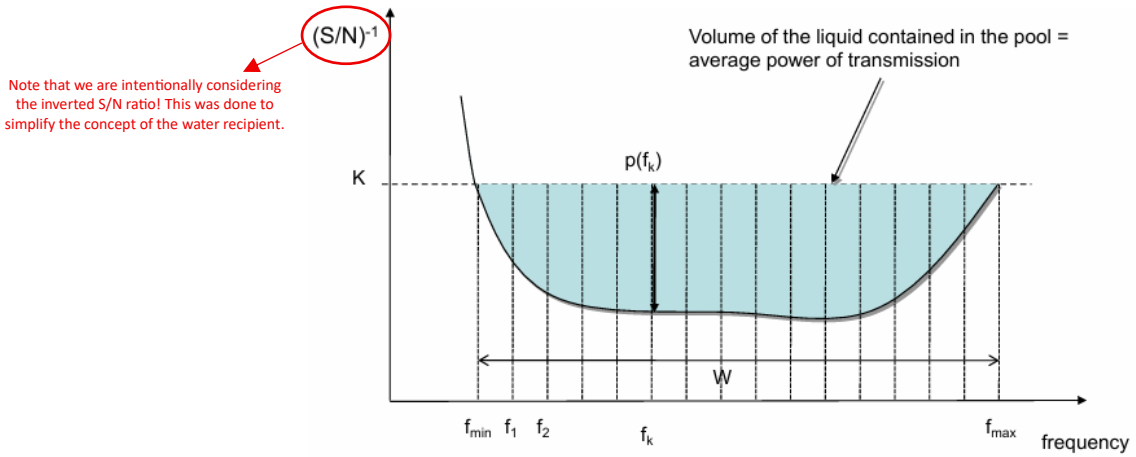
- **Upstream:** Using 25 allocated sub-channels:

$$\text{Bitrate}_{up} = \mathbf{25} \text{ channels} \times \mathbf{15} \text{ bit/s/Hz/channel} \times \mathbf{4} \text{ kHz} \approx \mathbf{1.5} \text{ Mbit/s}$$

- **Downstream:** Using 249 allocated sub-channels.

$$\text{Bitrate}_{down} = \mathbf{249} \text{ channels} \times \mathbf{15} \text{ bit/s/Hz/channel} \times \mathbf{4} \text{ kHz} \approx \mathbf{14.9} \text{ Mbit/s}$$

Water-Filling

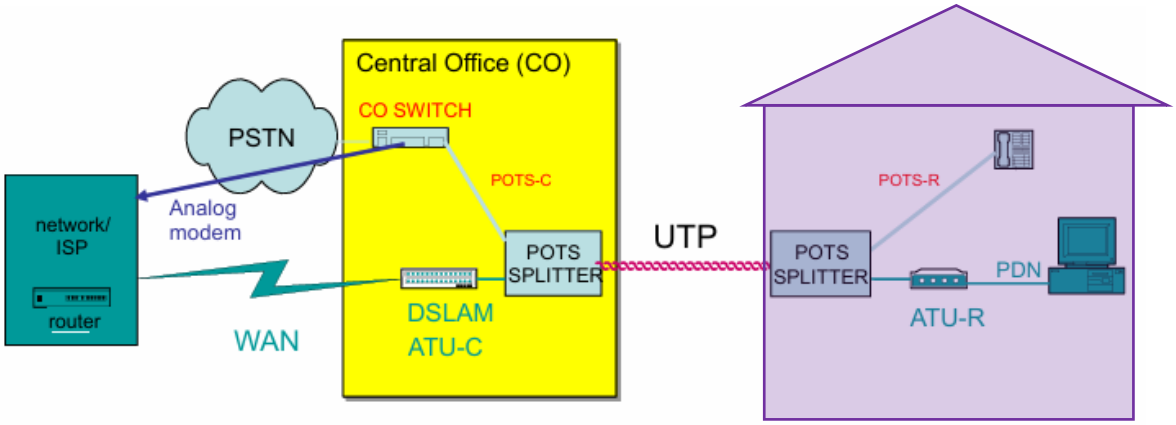


To optimize overall data rates across differing sub-bands, DMT implements an algorithm known as **Water-Filling**. The idea is that the signal transmission power across the spectrum is modeled like pouring water into a container. The bottom represents the noise level at each frequency f_k.

- **Frequencies with low noise (deep water)** receive higher transmission power, encoding more bits per symbol.
- **Frequencies with excessive noise (shallow water)** receive fewer bits or are left unused.

ADSL Architecture

The end-to-end ADSL architecture defines how user equipment interfaces with the copper local loop and connects back to the core transport network. The infrastructure is partitioned into the **Customer Premises** and the **Central Office (CO)**, linked by legacy **twisted-pair** cabling.



We have:

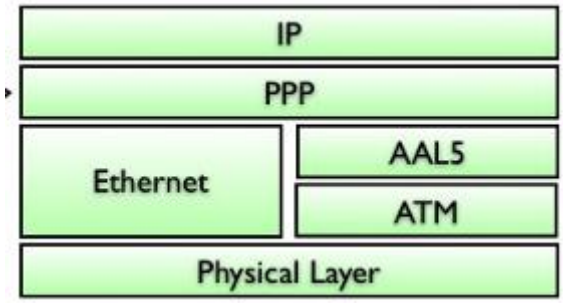
- **POTS Splitter:** Even though POTS and ADSL occupy distinct frequency ranges, telephone equipment can generate electrical interferences that leak into high frequencies (overlapping the digital ones). The splitter acts like a sort of filter to prevent these cross-band interferences.
- **ATU-R (ADSL Termination Unit – Remote):** Commonly referred to as the ADSL modem, it's located at the customer house. It terminates the digital subscriber line, executes Discrete Multi-Tone (DMT) modulation/demodulation, rate adaption, frequency allocation, echo cancellation, water-filling, and delivers IP traffic to the user's LAN.

- **UTP (Unshielded Twisted Pair):** The pre-existing copper cable plant originally installed for the telephone network. Each subscriber connects to the Central Office over a dedicated, unshared physical copper twisted pair.

- **Central Office (CO):** marks the boundary between the local access network and the provider's core network. It mirrors the subscriber-side hardware with carrier-grade equivalents:
- **POTS Splitter (CO Splitter):** Performs the reverse operation of the customer splitter. It separates low-frequency voice traffic and directs it into the traditional circuit-switched telephone network (PSTN/CO Switch), while high-frequency data is passed to the digital equipment.
- **ATU-C (ADSL Termination Unit – Central Office):** The central office counterpart to the ATU-R. It performs the same modem functions of before (modulation, tone negotiation, equalization).
- **DSLAM (Digital Subscriber Line Access Multiplexer):** Aggregates a large set of ATU-C interfaces, terminating up to several hundred subscriber lines in a single system. It multiplexes user traffic toward the core network/Internet (and demultiplexes incoming traffic back to individual lines), providing a centralized point for configuration, line diagnostics, and operator monitoring.

NOTE: Since ADSL was built upon the old telephone line, the central office architecture was not always like this. To adapt it for ADSL, modems (at both the central office and the user's home) and splitters were installed. However, the installation in the user's house is not a problem… but the installation in the Central Office of course is! It was necessary find space and installs hundreds of modems (1 for each user) in a space that was not designed for that!

ADSL Network Protocol - PPP

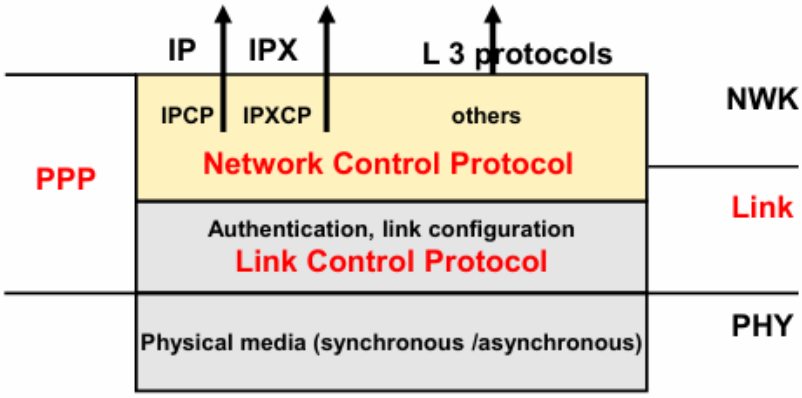


PPP (Point-to-Point Protocol) is a data link layer protocol designed for simple point-to-point links to transport packets between two direct peers. In the context of ADSL, it handles communication between the end user's modem/router and the ISP's Central Office equipment.

It's designed as a simplified version of the Ethernet protocol. However, unlike Ethernet (which was designed for multi-access local area networks), PPP was specifically built for dedicated links. It is used alongside ADSL because it provides critical carrier-grade access features:

- **Multiplexing:** PPP encapsulation allows multiple network-layer protocols (e.g., IPv4, IPv6) to run simultaneously over the same physical transmission link.
- **Access Control:** It supports essential subscriber-management operations, including user authentication (using protocols like PAP or CHAP), authorization, dynamic network parameter configuration, and integration with addressing mechanisms such as DHCP.

PPP Components

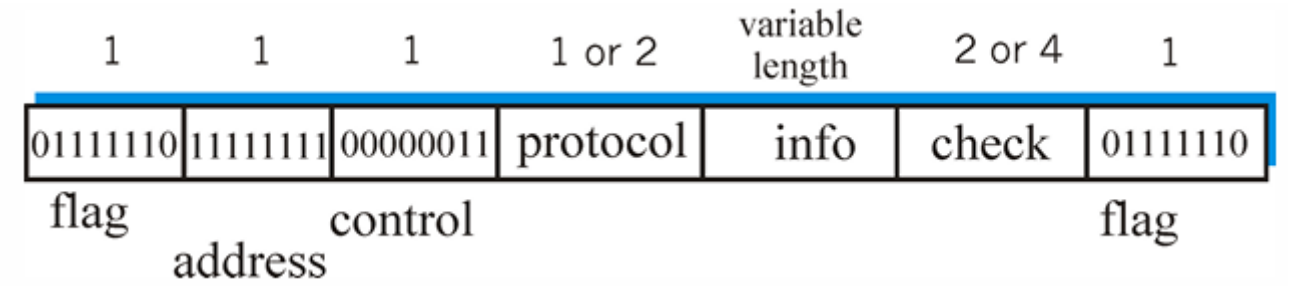


PPP relies on two main protocol sub-layers to set up and manage communication:

1. **LCP (Link Control Protocol):** Responsible for the **establishment**, **testing**, **configuration**, and **termination** of the point-to-point link. It negotiates physical and data link parameters, such as the maximum receive unit (MRU), encapsulation formats, link quality monitoring, and the choice of authentication method.
2. **NCP (Network Control Protocol):** Configures and manages the network-layer parameters once the physical link is established.

Because ADSL underlying transport networks evolved through different technologies, PPP frames can be incapsulated over ATM (and we name it as **PPoA**) or directly inside standard Ethernet frames (**PPoE**)

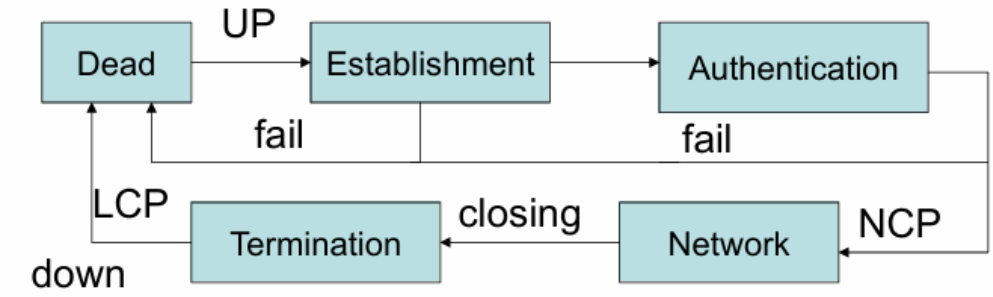
PPP Frame Structure



A standard PPP frame consists of several distinct fields arranged in order:

- **Flag (1 byte):** A well-known bit sequence placed at both the beginning and the end of the frame. It acts as a delimiter, allowing the receiver to recognize exactly where a frame starts and terminates.
- **Address (1 byte):** In the PPP case it is a dummy field, fixed to a sequence of all 1s. It happens because internet is a protocol designed to work in a network with several clients. PPP instead is used between 2 peers. We don't need to address anybody.
- **Control (1 byte):** Another legacy field typically set to 0x03 (indicating unnumbered information frames), ensuring compatibility with older HDLC standards.
- **Protocol (1 or 2 bytes):** Identifies the specific higher-layer protocol encapsulated inside the Information payload field. For example, it specifies whether the payload contains an IPv4 packet, an IPv6 packet, an LCP control packet, or an NCP management packet.
- **Information (Variable length):** Contains the actual datagram being transported across the link.
    - Its size can range from zero up to the **MRU** (**Maximum Receive Unit**).
    - The standard default MRU value is **1500 octets (bytes)**, which matches the default MTU of standard Ethernet.
    - The exact MRU can be dynamically renegotiated using LCP during link initialization.
    - If the data is shorter than required by the underlying physical layer, the Information field may be padded with extra bytes up to the agreed MRU size.
- **Checksum / FCS (Frame Check Sequence, 2 or 4 bytes):** A cyclic redundancy check (CRC) used by the receiving peer to detect transmission errors. If the calculated checksum does not match the received value, the corrupted frame is discarded.

PPP States

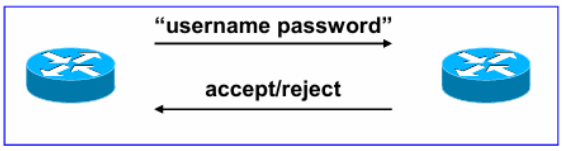


The operation of PPP follows a state-machine lifecycle divided into five sequential phases:

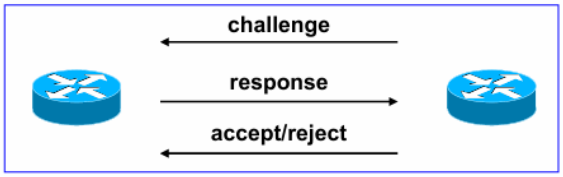
1. **Dead Phase**: The initial non-operational state of the link.

2. **Establishment Phase (LCP):** During this phase, the Link Control Protocol (LCP) sets up and configures the data link layer between the two peers. It negotiates frame sizes, compression, and error detection. The negotiation process consists of 2 messages:
    - **Configure-Request**: The initiating peer sends a message containing the desired operational parameters.
    - **ACK**: If the receiving peer accepts all parameters
    - **NAK**: If any parameter is unacceptable, suggesting alternative values to continue negotiation until an agreement is reached.

3. **Authentication Phase**: Once LCP completes link configuration, the connection moves to user verification before granting access to network resources. The authentication method depends on what was agreed upon during the Establishment phase:
    - **PAP (Password Authentication Protocol):** A basic two-way handshake where the client sends the username and password in clear text over the link. It is vulnerable to eavesdropping.



    - **CHAP (Challenge Handshake Authentication Protocol):** A more secure three-way handshake that avoids sending passwords in plain text (The authenticator sends a random string, named "challenge", to the user; The user calculates a hash using a one-way hashing function based on the shared secret password and the challenge string. The resulting hash is returned to the authenticator, which verifies the match. Because the password is never transmitted across the wire, CHAP is significantly more secure than PAP).



4. **Network Phase (NCP):** Configures the network parameters. The relevant Network Control Protocol (NCP) negotiates IP Addresses, DNS, gateway, etc…

5. **Termination Phase (and return to Dead):** The link is torn down. It could happen by user request, administrative disconnection by the ISP or physical loss/carrier failure.

VDSL (Very High-Speed Digital Subscriber Line)

To overcome the physical throughput limitations of ADSL, the copper access architecture evolved into VDSL (and VDSL2). The primary objective of VDSL is to achieve significantly higher bitrates over copper by altering two fundamental parameters:

1. **Shortening the Copper Loop:** In ADSL, the copper pair runs all the way from the Central Office to the user's home, often stretching several kilometers and causing high signal attenuation.

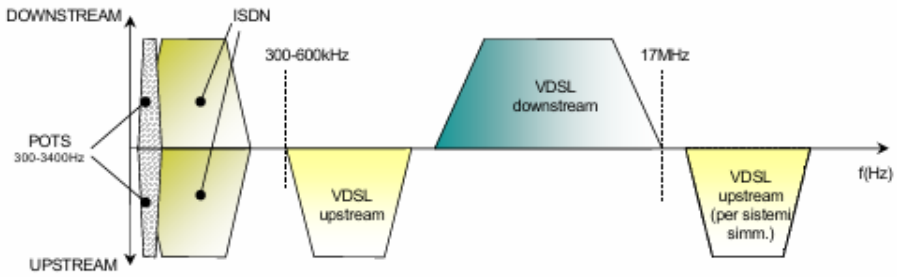
   VDSL mitigates this by shortening the copper loop: optical fiber is run from the Central Office to a street cabinet—an Optical Network Unit (ONU) or MSAN (Multi-Service Access Node)—installed close to the user premises. Only the final drop from the cabinet to the customer remains copper (this architecture is named FTTC – Fiber to the Cabinet).

2. **Expanding the Operational Bandwidth**: According to Shannon's Capacity theorem, increasing the bandwidth directly raises the bit rate. So, VDSL uses much higher frequency plans to achieve an higher bit rate.

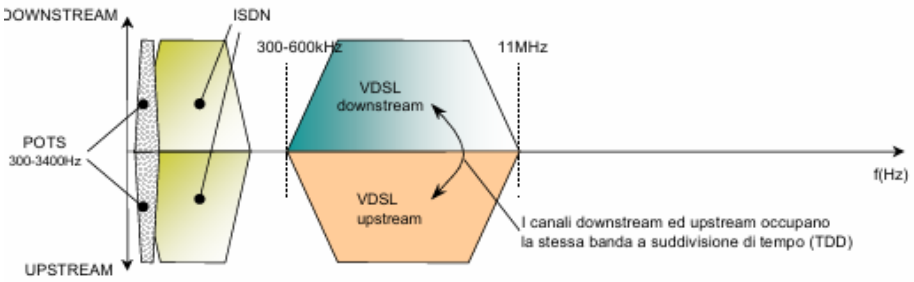
FDD vs. TDD

VDSL aims to maximize the bit rate, However, higher frequencies cause higher noise and interferences. So, to handle bidirectional transmission efficiently, VDSL can employ two duplexing methods:

- **FDD (Frequency Division Duplexing):** Upstream and downstream traffic are assigned to completely distinct frequency bands. Instead of having just one single band for upload and one for download (as in ADSL), VDSL commonly interleaves multiple upstream and downstream sub-bands across the spectrum to balance power spectral density and minimize crosstalk.

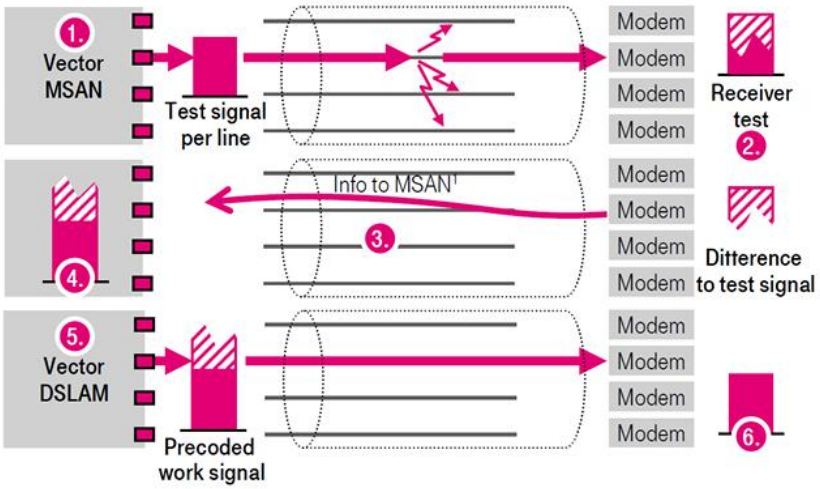


- **TDD (Time Division Duplexing):** Upstream and downstream share the exact same frequency band, but transmission time is sliced into distinct alternating time slots. The transmitter and receiver alternate their turns to send data, eliminating frequency fragmentation.



VDSL Vectoring

While shortening the copper loop reduces attenuation and enables high frequencies, bundling numerous active high-frequency lines into the same street cabinet binder causes severe electromagnetic coupling (crosstalk, especially FEXT). At VDSL frequencies, crosstalk becomes the single largest bottleneck limiting subscriber speeds.



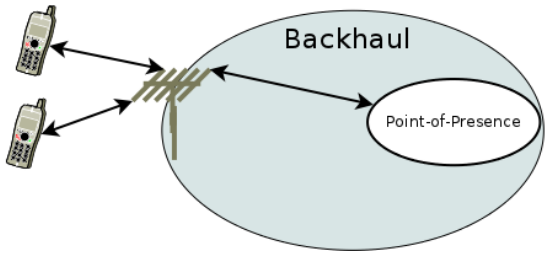
To cancel this crosstalk, VDSL implementations use Vectoring, which operates essentially like advanced noise cancellation across all lines in a binder:

1. **Interference Measurement (Pilot Signals):** The cabinet equipment (MSAN/DSLAM) sends known test signals (pilot sequences) across the lines.
2. **Feedback Loop**: The remote customer modems measure the error vectors and mutual crosstalk. Then sends these error reports back to the central unit.
3. **Pre-compensation (Anti-Noise Generation):** The vectoring processor inside the DSLAM calculates the complete crosstalk matrix of the binder. When transmitting downstream, it generates a precisely calculated "anti-phase" signal (anti-noise) and adds it to the transmitted data.
4. **Cancellation at Destination**: As the signals travel down the copper bundle, the physical crosstalk naturally cancels out the added anti-noise components. By the time the electrical waveform arrives at the end user's modem, the mutual interference is neutralized, leaving a clean, high-SNR signal that enables maximum bitrates.

Optical Fiber

As we already said before, due to the high demand for bandwidth and its improved performances, the optical fiber is progressively replacing copper in 2 different scenarios:

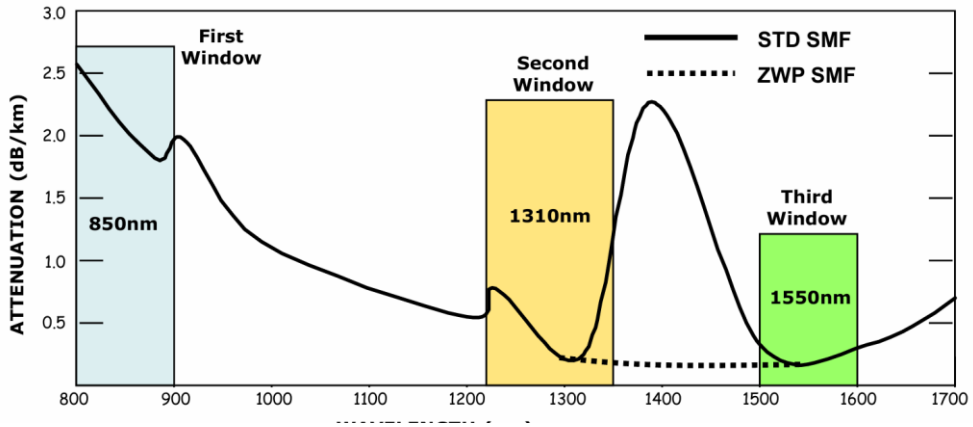
- **Bring the fiber as close as possible to the end user**, whether to the street cabinet (FTTC), the building (FTTB), or directly inside the home (FTTH), trying to eliminate as much as possible the high-attenuation caused by the copper.
- **Mobile Backhaul:** The backhaul is the intermediate segment linking the mobile access nodes to the core network. As wireless speeds climb into gigabit ranges, legacy copper backhaul becomes an intolerable bottleneck; modern radio access networks require direct optical fiber backhaul to transport aggregated mobile traffic with minimal latency.



Optical Fiber Characteristics and Transmission Windows

An optical fiber is basically a glass cable *(actually an ultra-pure cylindrical silica core surrounded by a cladding layer)* used to transmit light, typically using a laser, that encodes digital signals (bits).

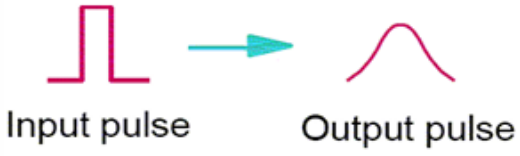
In optical networking, the carrier frequency is typically characterized by its **wavelength (λ),** measured in nanometers (nm).



In addition, the transmission is concentrated within specific spectral bands, known as **optical transmission windows**, where the glass exhibits minimal signal attenuation and favourable propagation characteristics:

- **Low and Flat Attenuation:** Within these transmissions, the optical power loss over distance is exceptionally low and relatively uniform across the passband. This allows optical signals to travel kilometers without intermediate amplification.
- **Massive Available Bandwidth:** In the optical frequency domain, these wavelength windows translate into tens of terahertz of usable spectrum, enabling enormous data capacities.

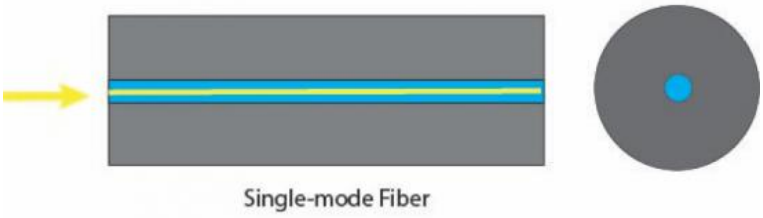
However, when a light pulse propagates along a fiber, it tends to spread out causing adjacent light pulses to widen and overlap at the receiver. This phenomenon is known as **dispersion**. A low-dispersion fiber preserves the sharp profile of the optical pulses, allowing higher modulation rates and longer transmission spans.



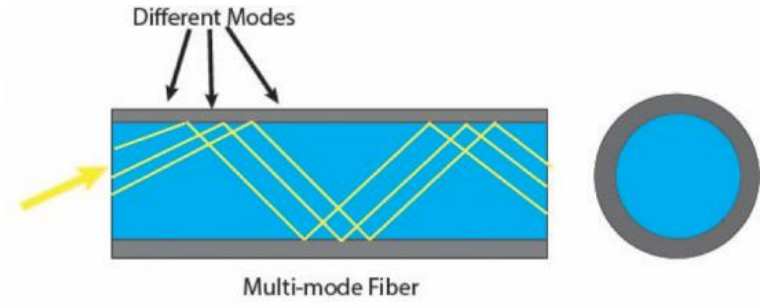
Fiber Propagation Modes

A critical technical distinction exists between the two fundamental types of optical fiber, which relates to physical ray paths rather than multiplexing multiple data streams:

- **Mono-Modal Fiber:** Features a very narrow core diameter permitting only one spatial ray to propagate down the glass. Because light takes only one single route, there is zero modal dispersion. Pulses do not spread due to differing path lengths. Consequently, Single-Mode Fiber offers the highest performance, supporting the highest data rates and transmission distances of tens to hundreds of kilometers. It is the standard media used for telecommunication backbones, submarine links, and modern FTTH access networks.



- **Multi-Modal Fiber:** Features a wider core diameter, allowing multiple light rays contemporary. Because rays traversing higher-order modes travel longer physical distances than axial rays, they arrive at the receiver at slightly different times. This introduces severe modal dispersion, which broadens optical pulses quickly over distance. As a result, Multi-Mode Fiber is strictly limited to short-range applications (typically under 500 meters, such as local connections within a data center or building campus). It is not used for long-distance telecommunication or regional access networks.



Note: Transmitting multiple simultaneous data streams over a single fiber is achieved by using different optical wavelengths or "colors" of light—a technique known as WDM (Wavelength Division Multiplexing)—which is implemented on Single-Mode Fiber, not by using Multi-Mode Fiber.

Amplification

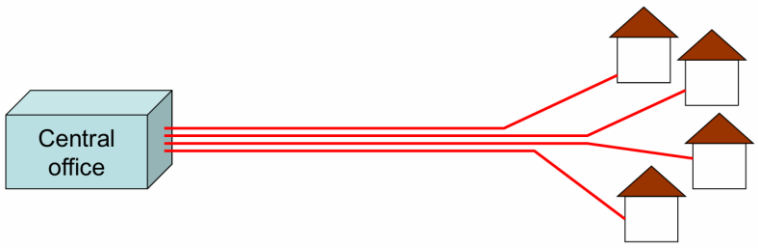
Over long distances (such as continental networks or transoceanic submarine cables) light eventually loses power due to the attenuation. So, instead of converting the optical signal into an electrical signal to regenerate and retransmit it (which adds hardware complexity, latency, and throughput caps), networks employ **in-line optical amplifiers**, placed directly into the fiber line at periodic intervals.

When the incoming weak data signal passes through the energized fiber segment, it stimulates photon emission at the exact same wavelength and phase, amplifying the optical signal directly in the optical domain without electronic conversion.

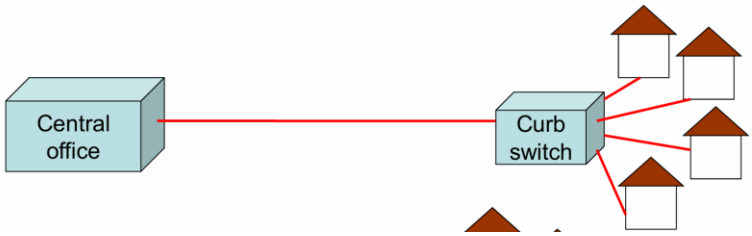
Passive Optical Network (PON)

(As we have already partially seen) Optical fiber access networks (FTTx) are deployed according to three primary architectural models, each presenting different trade-offs between costs, complexity, and infrastructure sharing:

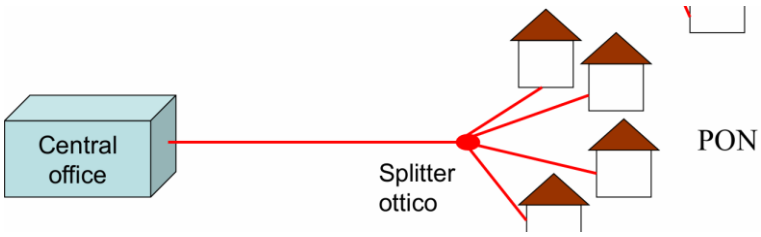
1. **Point-to-Point (P2P / Dedicated Fiber):** A dedicated, unshared optical fiber runs directly from the Central Office to each individual subscriber. While this model guarantees maximum dedicated bandwidth and avoids optical sharing bottlenecks, it carries prohibitive deployment costs (mainly due to the digging cost).



2. **Active Optical Network (AON / Active Curb Switch):** Optical fiber runs from the Central Office to an intermediate street-level curb switch that converts the optical signal into a electrical one. Then re-converts the signal back into an optical to send it directly to the end users. This configuration is good, but requires space and active energy to power the active node/switch.



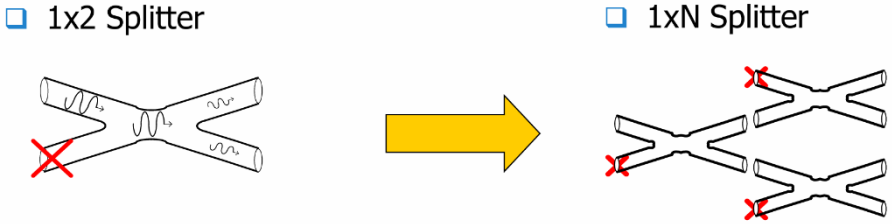
3. **Passive Optical Network (PON):** The dominant and most cost-effective architecture for modern FTTH deployments. PON replaces active street switches with completely **passive optical splitters**. Because these components require no electrical power, street-level active electronics are eliminated entirely.



Passive Splitter

An optical splitter is an unpowered optical component, typically manufactured by fusing optical fibers together (Fused Biconical Taper) or using planar lightwave circuits (PLC).

- **Point-to-Multipoint Splitting:** To "divide" and propagate the signal from one sender to multiple receivers, the passive splitter closes one entry of the glass. In this way, the light enter the splitter on a single fiber from the Central Office, and then it is physically divided into multiple output branches to serve multiple customer lines.



- **Optical Splitting Loss:** However, each split introduces an inherent, unavoidable physical power loss. As the number divisions increases, the cumulative attenuation limits the maximum number of subscribers that can share a single feeder fiber.

The important fact is that: even with the signal loss it still provides a very high quality. And, differently from before where the maximum distance between the end user and the central office was just 1.5 kms with the fiber we can reach even tens of kilometres.

PON Network Elements

PON operates as a point-to-multipoint optical distribution network (ODN) over a single shared strand of single-mode fiber, utilizing two main terminal elements:

- **OLT (Optical Line Terminal):** Located at the provider's Central Office, it acts as the head-end engine, converting core network IP/Ethernet traffic into optical signals and orchestrating downstream and upstream transmissions across all connected endpoints.
- **ONT (Optical Network Terminal) or ONU (Optical Network Unit):** Located at the user side, it terminates the optical fiber, converts optical signals back into standard electrical Ethernet signals, and presents user-facing interfaces (LAN ports, Wi-Fi, voice).

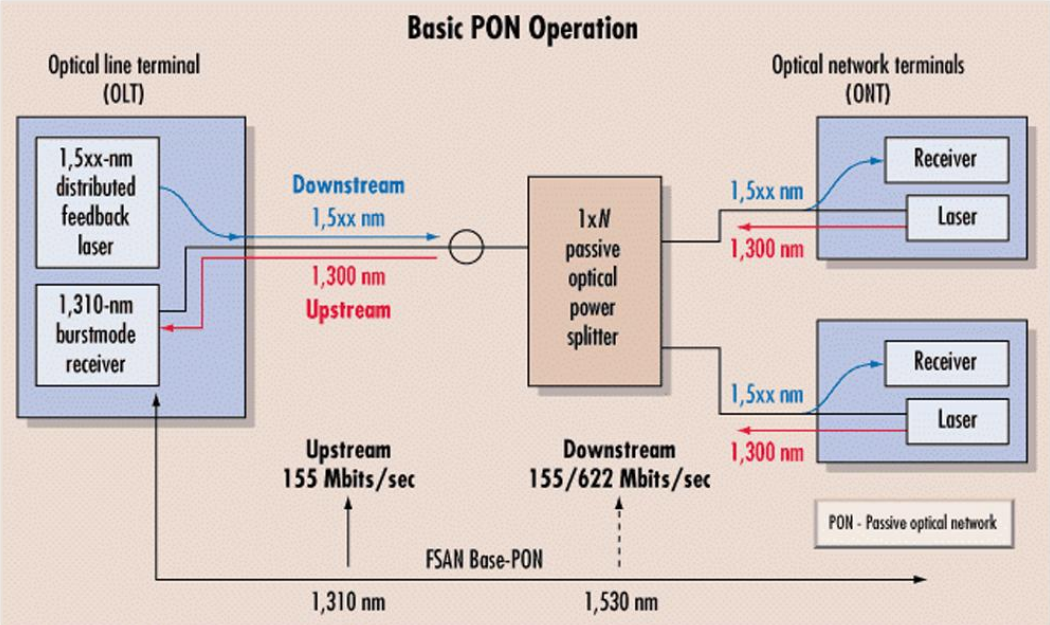


Figure 1. A basic passive-optical-network operation enables service to as many as 32 optical-network terminals (ONTs). Each ONT, in turn, can be connected to multiple subscribers, making fiber more affordable for access networks.

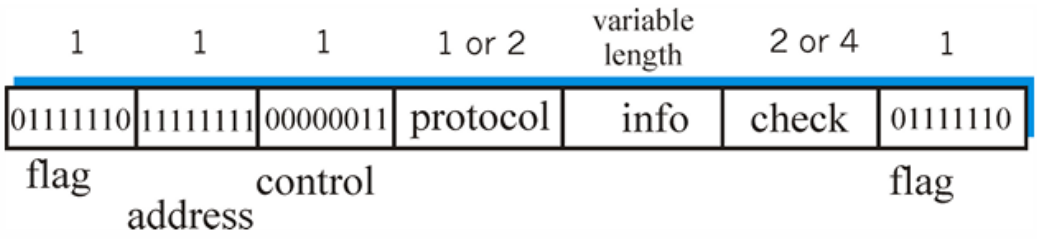
ATM vs EPON

These things we just explored are the “physical way” (Layer 1) in which the transmission happen. On top of this we need a Layer 2 protocol:

- **APON / BPON (ATM-based PON):** Early PON standards were built on **ATM** (Asynchronous Transfer Mode), which breaks all traffic into small, fixed-size cells. However, it introduced high protocol overhead and significant complexity when carrying native IP traffic.
- **EPON (Ethernet PON):** As Ethernet became the universal standard for local and metropolitan networks, modern PON architectures transitioned to Ethernet framing, eliminating the overhead of ATM cell segmentation.
- **GPON (Gigabit PON): E**ncapsulates variable-length Ethernet frames using a dedicated framing method (GEM - GPON Encapsulation Method).

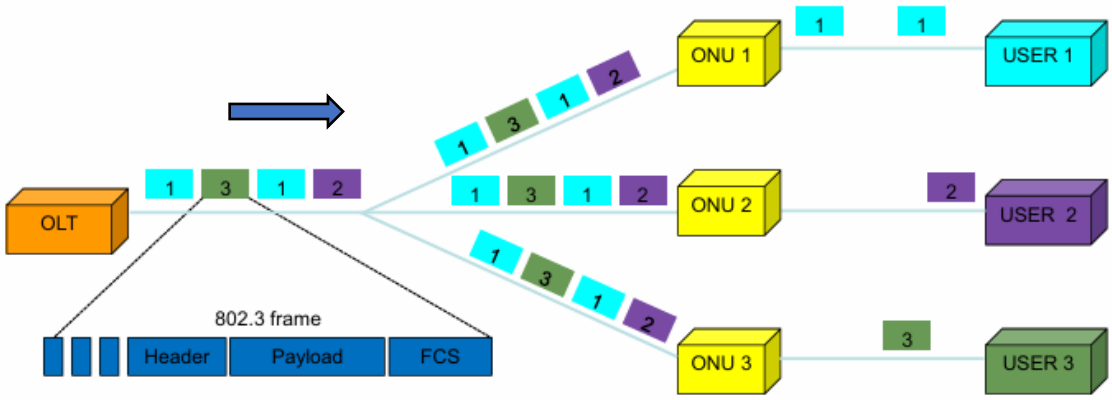
PON Downstream

Assuming that we are using EPON, we have already seen the Ethernet Frame structure. However, we saw it with for the Point-to-Point Protocol (PPP), where the internet address field was just "dummy" (set to all 1s for convention).



In the PON case, instead, the internet address field is used to distinguishes different destinations and cannot set randomly. However, this new introduction of addresses also introduced new challenges, forcing the engineers to modify the Ethernet protocol to adapt the changes.

A second issue was related to the user's privacy: since we have a single channel for all communications, each user could read the content of the others. So, we must encrypt the content of the message



TDM vs TDMA

In the **downstream** direction, the architecture has only a single transmitter (the OLT at the Central Office). Since there is only one central device controls the entire transmission, optical collisions are physically impossible.

So, the OLT simply takes the data packets destined for different users and places them sequentially along a continuous, uninterrupted stream of light. This mechanism is standard **Time Division Multiplexing (TDM)**. The passive splitter downstream does not make routing decisions; it merely acts as a physical broadcast device, replicating this identical light stream to all connected Optical Network Terminals (ONTs). Each user device receives the complete stream, filters out traffic intended for other subscribers using Layer 2 destination addresses, and decrypts only its assigned payload.

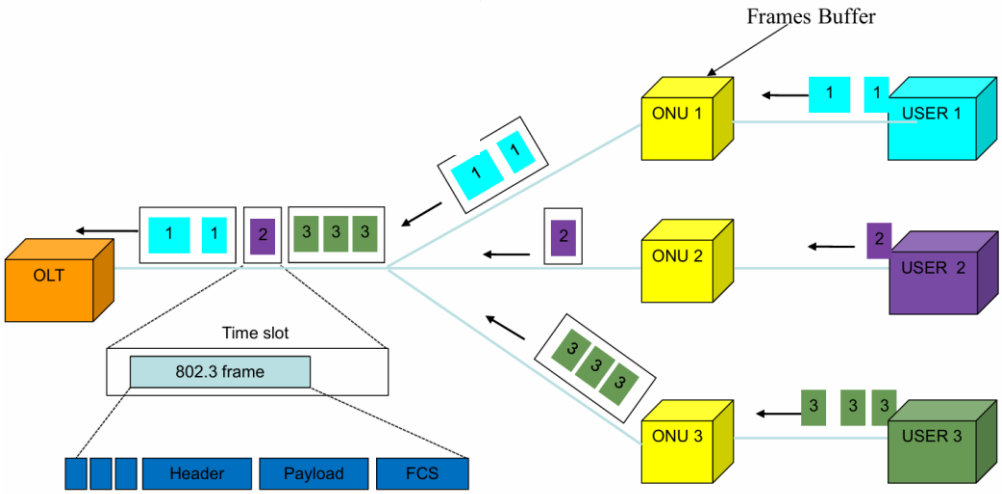
NOTE: In contrast, the **upstream** direction presents a classic multiple-access problem. Multiple individual ONTs send data to a single receiver (the OLT). Consequently, if two or more ONTs turn on their lasers at the same moment, their optical pulses would overlap, causing unrecoverable optical collisions. To prevent this, the upstream path requires **Time Division Multiple Access (TDMA)** (We'll see later)

PON Upstream

In traditional Ethernet networks, collisions are avoided using **CSMA (Carrier Sense Multiple Access)**, where devices listen to the channel before transmitting. However, we cannot use CSMA in a PON architecture. Because the optical splitter is completely passive, it cannot make any routing decisions or coordinate traffic. Furthermore, the optical path between users is isolated (light from one user travels only upstream toward the central office) meaning end users cannot "hear" each other transmitting.

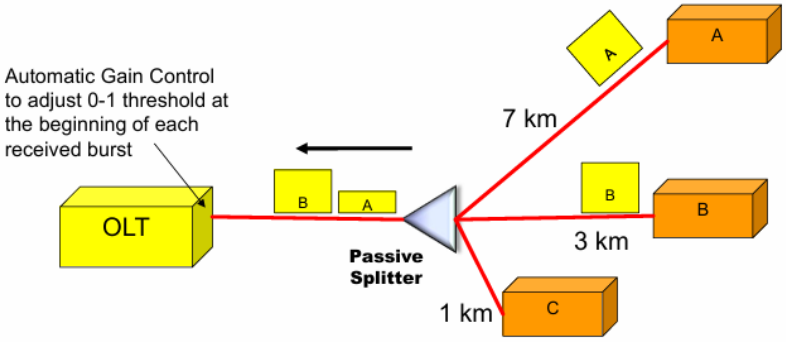
To solve this problem, the network must prevent users from transmitting at the same time. This is achieved by scheduling transmission turns through **TDMA (Time Division Multiple Access)**. In this scheme, the Optical Line Terminal (OLT) at the Central Office acts as the master coordinator: it assigns specific time slots to each user through control messages.

In this way, data arrives organized in a single stream where packets cannot collide. However, this technique leaves a small amount of bandwidth unused, as the network must insert small time gaps (**guard times**) between slots to ensure that messages from different users do not overlap.



Automatic Gain Control (AGC)

Another challenge is related to the distance and the signal power of the different users. Because subscribers are located at varying distances from the Central Office, a node that is farther away will suffer higher signal attenuation than a closer one. This makes it difficult for the receiver to recover synchronization and read the data cleanly, since incoming bursts arrive with wildly different power levels.



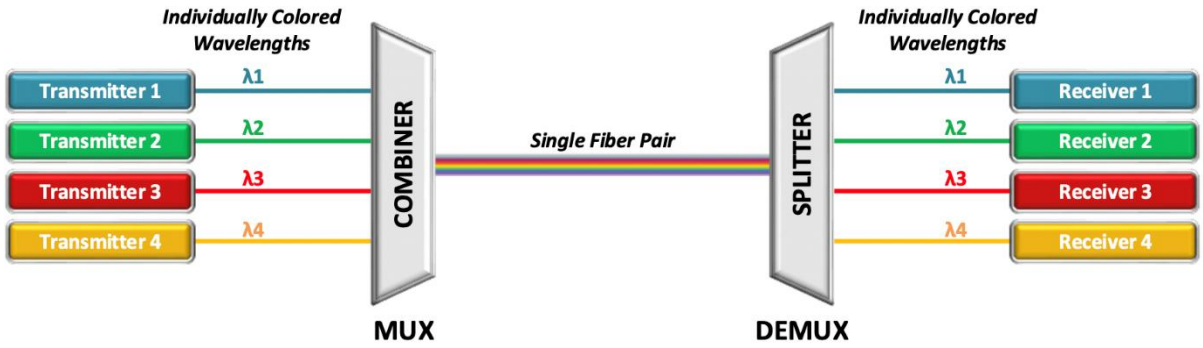
To handle this, the receiver uses Automatic Gain Control (AGC), which dynamically adapts to the incoming signal power from each user burst. The OLT also measures the distance (propagation delay) of each user to equalize transmission timing across the network. Then, it tells them the amount of power they should use to send the message.

WDM-PON (Wavelength Division Multiplexing)

In a standard PON, all subscribers share the exact same downstream wavelength using time slots (TDM) and the same upstream wavelength using TDMA.

WDM-PON overcomes this shared bandwidth limit by exploiting multiple optical wavelengths, transmitting data across different "colors" of light over the same optical fiber simultaneously.

Instead of dividing time, the network assigns a dedicated wavelength (color) to each subscriber. In the downstream direction, all these different wavelengths travel together across the shared feeder fiber. When the light reaches the distribution point, instead of a simple power splitter that duplicates the entire signal to everyone, the network uses an **Optical Router** (a passive wavelength demultiplexer, historically based on tiny reflective mirrors and today built using optical prisms).



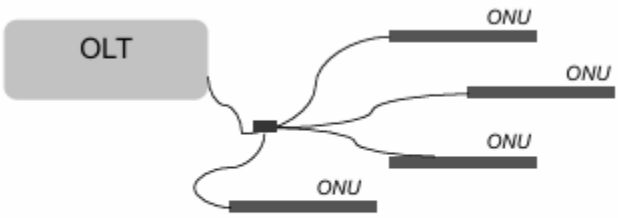
By replacing time sharing with wavelength separation, WDM-PON provides key advantages:

- **Dedicated Bandwidth:** Each user receives the full capacity of an entire optical wavelength without competing for time slots.
- **Inherent Privacy:** Downstream traffic is physically filtered at the optical router, so a customer's ONT receives only its assigned wavelength, eliminating the need to encrypt a shared broadcast stream.
- **Simplified Protocol Stack:** Because each customer has a dedicated optical channel, complex scheduling algorithms (like TDMA ranging and guard times) are no longer strictly necessary.

EPON Configurations

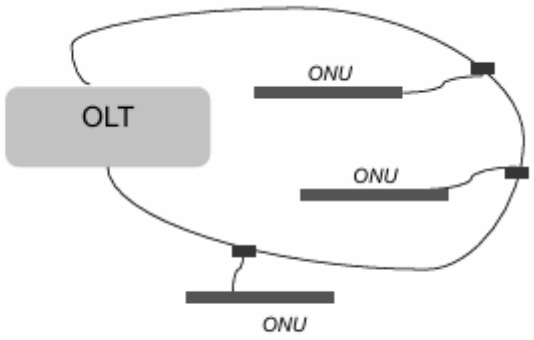
In an Ethernet Passive Optical Network (EPON) we can combine the OLT and the ONUS in different topologies:

- **1. Tree Topology (Standard Point-to-Multipoint)**: A single feeder fiber runs from the OLT to a centralized passive optical splitter which distributes individual drop fibers to each ONU. It maximizes fiber sharing, minimizes deployment costs, and is simple to manage. However, we have a single point of exists along the primary feeder fiber.



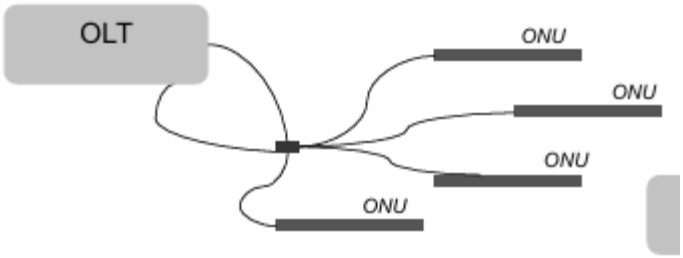
1) Tree Topology

- **2. Ring Topology**: A continuous fiber loop originates and terminates at the OLT. Multiple tap splitters are placed along the perimeter of the ring to drop signals to local ONUs. It provides physical path redundancy and self-healing capability. If a fiber break occurs anywhere along the loop, traffic can be redirected or counter-propagated from the opposite direction. However, the signal attenuation varies along the ring and requires directional splitters.



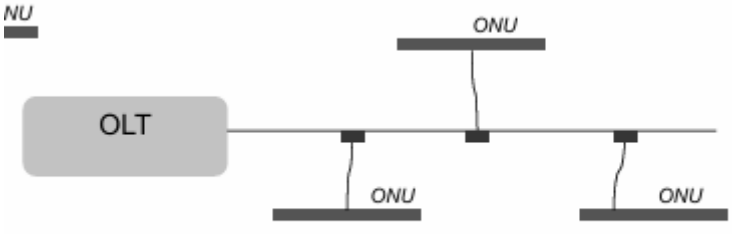
(2) Ring Topology

- **3. Tree with Redundant Trunk:** Uses a standard tree architecture, but with **two distinct feeder fibers** running from two separate OLT ports. The distribution drop fibers to the ONUs remain single-run lines. Partially solves the single point of failure. If the primary trunk fiber is cut, the OLT automatically switches transmission to the backup link with minimal downtime. However, individual drops to ONUs still represent single points of failure for respective end-users.



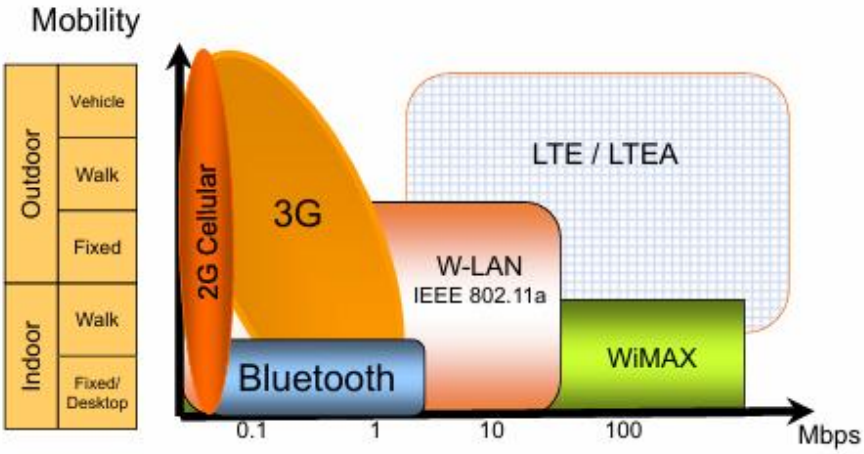
(3) Tree with Redundant Trunk

- **4. Bus Topology (Daisy-Chain)**: A single backbone fiber extends linearly from the OLT, using inline splitters at sequential points along the line to link each ONU. It is well-suited for linear layouts (e.g., highways, railways, etc...) but it is not very used. Each inline tap introduces insertion loss, meaning downstream ONUs receive significantly less optical power than upstream ONUs. In addition, it introduces an enormous point of failure: a fiber cut near the OLT disconnects all downstream ONUs along the bus.



(4) Bus Topology

Wireless Access



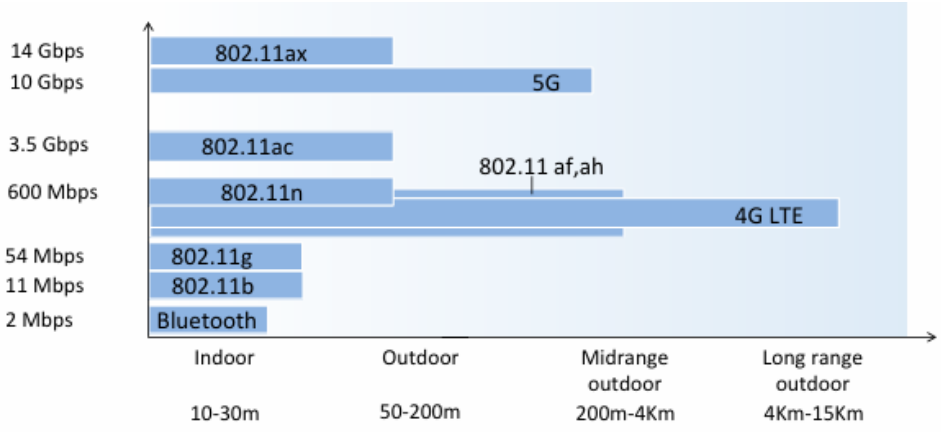
Wireless access allows us to connect to a data network over radio frequency channel that does not require any cable. Wireless access technologies are differentiated primarily by:

- **Different Bitrate Transmissions:** Different standards operate across varying frequency bands, offering distinct transmission capacities.
- **Coverage Scope (Indoor vs. Outdoor):**
    - **Indoor systems** (e.g., standard Wi-Fi, Bluetooth) cover short distances (typically 10 to 100 meters) and are optimized for high data density within confined spaces.
    - **Outdoor systems** (e.g., 4G/5G cellular networks) provide wide-area coverage (macro-cells extending up to several kilometers).
- **Handover (or Handoff):** The ability of the network to maintain an uninterrupted data session while the user moves rapidly (e.g., inside a vehicle or high-speed train), changing from a base station/cellular cell to another.

Comparison of Wireless Technologies

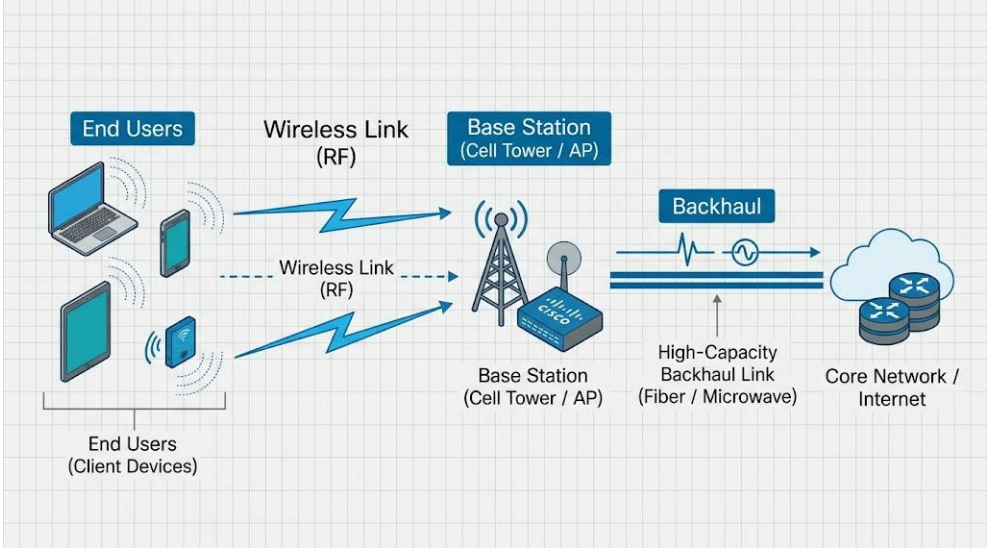
Wireless technologies differ primarily along two **trade-offs: coverage area and data rate**.

Higher carrier frequencies offer larger channel bandwidths, which translates directly into higher data rates (following Shannon's law). However, higher frequencies suffer from severe signal attenuation, higher path loss, and poor penetration through physical obstacles (such as walls, foliage, or even atmospheric moisture). Lower frequencies propagate much farther and penetrate obstacles more easily, but offer narrower bandwidths and lower maximum bitrates.



- **4G LTE:** Offers the widest outdoor coverage, operating in low frequency bands that travel long distances and easily pass through physical barriers. However, it “only” offers an upper bound of roughly 600 MBps).
- **5G:** Offers mid-range to short-range outdoor coverage (less that 4G!!) but transmits on much higher frequency ranges that ensures much higher bit rates, up to 10 GBps. However, transmitting on higher frequencies, the signal is easily blocked or absorbed by physical obstacles, even very thin materials. Because the signal cannot travel as far, 5G cannot achieve the wide single-antenna coverage of 4G. To achieve these massive data rates and maintain continuous coverage, 5G networks require cell densification, deploying a significantly higher number of antennas to reach the end users.
- **Wi-Fi:** Focuses on short-range indoor coverage (10 to 100 meters) ensuring high data rates within a localized area, but lacks broad outdoor mobility by default.

Infrastructure Mode



In infrastructure mode, all wireless host devices communicate exclusively through a centralized architecture that relies on antennas (also named Base Stations). Host devices never exchange packets directly over the air. Traffic must pass through the base station first. The main Network Elements are:

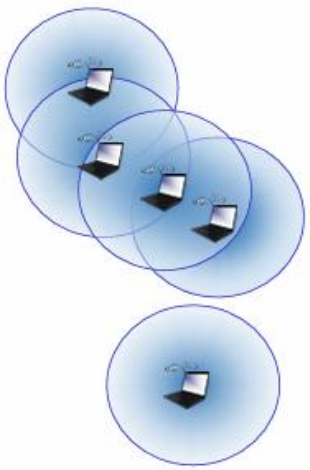
- **End Users / Wireless Hosts:** Mobile devices (smartphones, tablets) or fixed stations (laptops, IoT nodes) running client applications.
- **Wireless Link:** The physical radio channel connecting the host to the base station, governed by a medium access control (MAC) protocol.
- **Base Station:** The wireless transceiver responsible for coordinating channel access, scheduling transmissions, and relaying traffic between the wireless medium and the wired infrastructure. Every communication must pass through the base station first.
- **Backhaul:** The intermediate network segment (historically legacy copper, now almost universally optical fiber) that links local base stations back to the high-capacity core backbone.

Mobility and Handover: When a mobile user moves outside the coverage area of one base station, the network executes a **handover** (or **handoff**) procedure, dynamically migrating the active session to an adjacent base station without dropping the connection.

NOTE: Cellular systems are designed to offer mobility and centralized handovers from the beginning. On the contrary, the Wi-Fi standard was originally deployed without mobility features. It was added later (e.g., university campus networks allow us to move from a access points to another without requiring any additional login).

Ad Hoc Mode

In an ad hoc network, there is **no infrastructure needed…** no centralized controller, and no base station. (e.g., mobile hotspots or Bluetooth). In a ad hoc mode, nodes communicate directly as peers over the shared wireless medium.



Exploiting these peer-to-peer ad hoc communication we can create real networks without any infrastructure. This method is known as **Multi-Hop Relaying.** In this scenario, intermediate participating devices act as temporary routers, forwarding packets across multiple wireless hops to reach the destination.

- **Advantages:**
  - **Rapid Deployment:** Operates immediately without pre-existing cabling, cellular towers, or access points.
  - **Direct Communication and Low Latency:** Proximity-based peer-to-peer links avoid the round-trip overhead of routing traffic through a distant base station or central core.

- **Disadvantages:**
  - **Dynamic Topology Management:** Node movement causes links to break and reform continuously, requiring complex, self-healing ad hoc routing protocols.
  - **Redundancy and Reliability:** The network relies on peer availability; if critical intermediate forwarding nodes depart, the route breaks.
  - **Energy Constraints:** In general, portable ad hoc networks (smartphones, sensor networks), nodes rely on finite internal batteries, making routing overhead costly.

FOCUS: Appealing Applications

- **MANET (Mobile Ad Hoc Network):** A general topology of autonomous mobile devices moving unpredictably.
- **VANET (Vehicular Ad Hoc Network):** A specialized class of MANET where moving vehicles act as network nodes to exchange safety alerts, traffic conditions, and sensor data (V2V - Vehicle-to-Vehicle communication). Examples of this field include the self-driving cars communicating each other. This branch of ad hoc networks does not suffer from energy limitations because equipment is powered by the vehicle's electrical system.

Infrastructure vs Ad-Hoc Mode

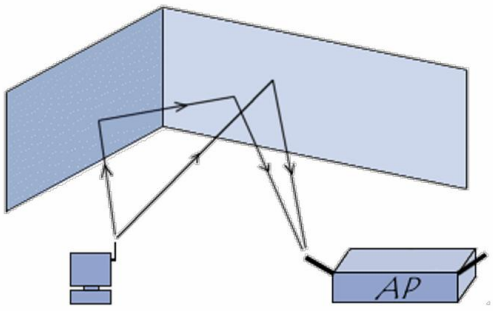
	single hop	multiple hops
infrastructure (e.g., APs)	host connects to base station (WiFi, cellular) which connects to larger Internet	host may have to relay through several wireless nodes to connect to larger Internet: *mesh net*
no infrastructure	no base station, no connection to larger Internet (Bluetooth, ad hoc nets)	no base station, no connection to larger Internet. May have to relay to reach other a given wireless node MANET, VANET

NOTE: This could be a question in the written exam!!!

Wireless Limitations

Compared to wired media (copper/fiber) where electromagnetic or optical signals are contained within a guided physical channel (the wire), wireless communication transmits data through the air. Consequently, the signal spreads all around and does not follow any guide. This fundamental difference introduces three main limitations that significantly impact link reliability and throughput:

- **Path Loss and Free-Space Attenuation:** Radio frequency (RF) signals attenuate much more than the communications via wire, causing the signal power density to decrease rapidly as a function of the distance. To quantify the area within which transmission can occur without issues, the "range" (or "radius of action") is defined; this represents the area in which the message can still be decoded.
- **Multipath Propagation:** Multipath is a physical impairment **exclusive to wireless transmissions**. When an unguided radio wave encounters obstacles (e.g., walls, buildings, terrain, or moving vehicles), the wave hits the objects and gets reflected away. As a consequence, the transmitted signal could reach the receiver along multiple distinct trajectories, each having a different physical length, attenuation level, and propagation delay.



- **Interferences:** Electromagnetic interference occurs whenever two or more devices transmit on the same time within the same frequency band. As we said before, due to the multipath problem, a single destination receives multiple signals from the same sender, but arriving from different directions, which may overlap and cause interference. Similarly, it is possible for different transmitters to use the same frequency band, resulting in overlap.

  (Note that mobile network does not interfere with wifi signals since they are on different frequency bands. Sometimes, 2.4GHz wifi could interfere with Bluetooth since they share the same frequences).

The combination of these problems makes wireless link considerably more complex than managing equivalent wired connections.

Signal-to-Noise Ratio (SNR) and Bit Error Rate (BER)

In wireless transmissions, the reliability and physical throughput of a link are fundamentally governed by the relationship between the **Signal-to-Noise Ratio (SNR)** and the **Bit Error Rate (BER)**:

- **Bit Error Rate (BER):** The fraction or percentage of transmitted bits that arrive corrupted at the destination due to channel noise, attenuation, or interference.
- **Signal-to-Noise Ratio (SNR):** The ratio of received target signal power to the background noise power. A high SNR indicates that the signal dominates the noise, allowing the receiver to reliably detect and decode the transmitted bits. A low SNR means the signal is drowned out by noise, leading to frequent bit detection errors.

NOTE: Increasing the transmission power raises the received SNR, which directly reduces the BER.

Channel Quality Measurements

Before 4G, channel quality and resource allocation were originally based on generalized averages across all active users in a cell. This static approach was inefficient: users near the cell edge experienced poor connections, while users close to the base station operated below their maximum potential capacity.

To improve this mechanism, from the 4th generation onward, networks introduced fine-grained Channel Quality Measurements. Each user device continuously measures its channel conditions (SNR, interference levels) and reports these metrics back to the base station. The base station then uses Adaptive Modulation to dynamically assign the optimal modulation order, coding rate, and power level to each user in real time.

Hidden Terminal Problem

Another important situation that could happen in the wireless networks is the **Hidden Terminal Problem**.

This problem occurs when two transmitting devices (e.g., node $\boldsymbol{B}$ and node $\boldsymbol{C}$) are outside of each other's radio transmission range, meaning they cannot hear or sense each other's transmissions to detect collisions, BUT they both share a common destination node (e.g., node $\boldsymbol{A}$, typically an Access Point or intermediate station) that falls within the coverage areas of both senders.

In this scenario, both transmissions arrive at node $\boldsymbol{A}$ at the same time, causing a collision. Consequently, node $\boldsymbol{A}$ cannot decode any of the 2 messages.

The exact same situation occurs even when nodes $\boldsymbol{B}$ and $\boldsymbol{C}$ are physically close enough to hear each other, but an intermediate physical obstacle (such as a thick wall or a building) blocks or severely attenuates their radio signals, preventing mutual carrier sensing.



How do we solve this? CSMA/CA

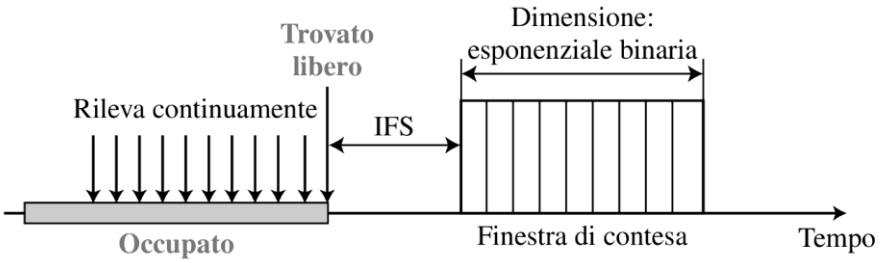
CSMA/CA

The CSMA/CA protocol is a variant of CSMA (Carrier Sense Multiple Access) that involves, like all Carrier Sense protocols, listening to the channel before starting to transmit data. In particular, Collision Avoidance allows avoiding collisions by adopting Interframe Spaces (IFS) which require devices to wait for a specific time interval, longer or shorter depending on priority, even when the channel is idle. The main IFSs are:

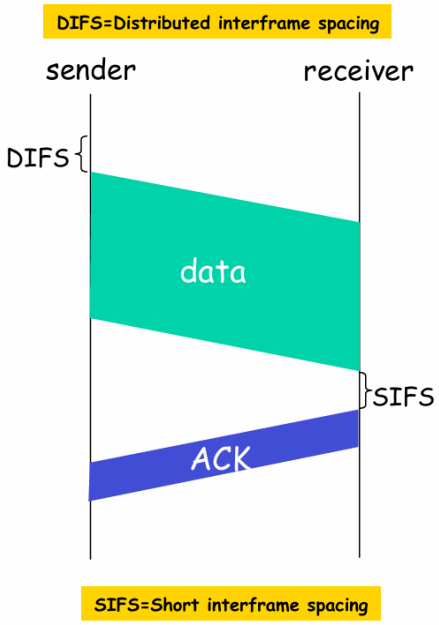
- **SIFS (Short InterFrame Space):** The shortest of all. Used for frames that must be transmitted without delay, with maximum priority. Typical use cases include sending an ACK (after receiving a frame) or sending a CTS (after an RTS; we will see later what these mean).
- **DIFS (Distributed InterFrame Space):** Used to start a new data transmission. It is longer than SIFS and is typically used when a device wants to transmit a new data frame and finds the channel idle. It is the standard for channel access in CSMA/CA.

Protocol:

1. **Carrier Sense (channel listening):** Like all variants of CSMA, it begins by listening to the channel. If the channel is busy, it continues listening until it becomes idle.
2. **Waiting for a DIFS duration**: Even if the channel has been sensed as idle, it is necessary to wait for a fixed interval before being allowed to transmit. This ensures that response frames (such as ACK or CTS, which must wait for a SIFS < DIFS) can be sent with priority over new transmissions.
3. **Random Backoff**: To prevent multiple devices from transmitting immediately after the DIFS, each node chooses a random number of time slots to wait (backoff) in the interval [0, CW-1], where CW = Contention Window. During the backoff, the device is not transmitting, so it can listen to the channel. Specifically, at each time step (Slot Time) of the interval (backoff), the device senses the channel to check whether someone else is transmitting.



   If no one is transmitting, it decrements the backoff counter by one. If another device starts transmitting instead, the backoff is paused and resumes once the other device has finished. (In particular, once the other device finishes, it is necessary to wait for a DIFS before restarting the decrement of the backoff timer). This allows handling cases where multiple devices entered the channel after the DIFS and risk transmitting on top of one another.
4. **Transmission**: When the backoff timer reaches 0, the frame is transmitted immediately.
5. **ACK**: After transmitting, the device waits for an acknowledgment message (ACK) from the receiver. If it does not receive it (due to a collision or error), transmission is retried after recalculating a backoff with a larger window (usually it grows as exponential of 2)

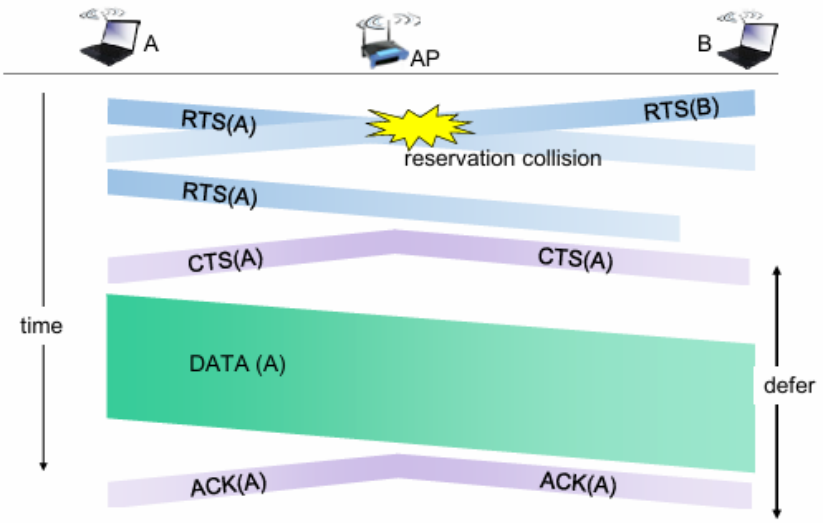


CSMA/CA with RTS and CTS

However, despite the use of IFSs that make it possible to avoid collisions rather than detect them, the CSMA/CA protocol still does NOT solve the hidden terminal problem. It is therefore necessary to extend the protocol by using a channel reservation mechanism through messages:

- **Request-to-send (RTS):** sent by the sender to request access to the channel. It includes the estimated duration of the transmission (frame + ACK). It notifies the nodes near the sender.
- **Clear-to-send (CTS):** sent by the receiver if the channel is idle, to grant permission to transmit. It notifies the nodes near the receiver.

This initial exchange allows all nodes in the vicinity of both the sender and the receiver to know that a transmission is underway and avoid interfering.

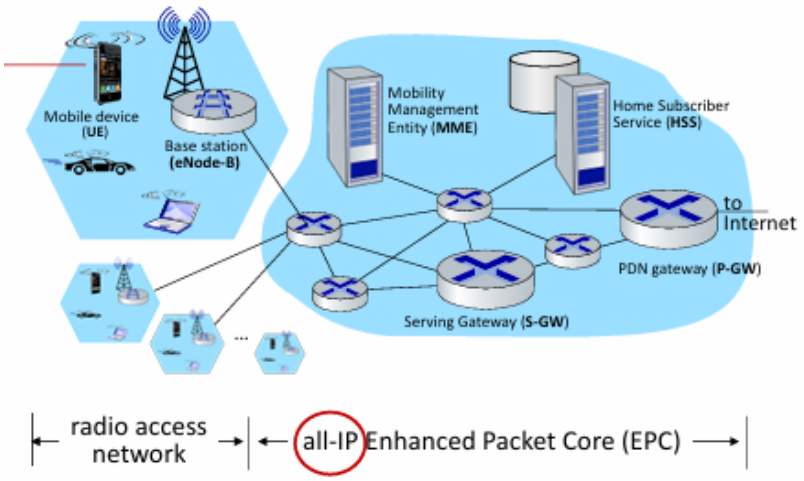


4G and 5G

Cellular architectures (4G LTE and 5G) differ fundamentally from local wireless networks like Wi-Fi in their core design goals:

- **Native Mobility:** Cellular networks were architected from inception to support high-speed user mobility without session interruption, coordinating handovers seamlessly across base stations.
- **Licensed vs. Unlicensed Spectrum:**
    - **Cellular (Licensed):** Telecom operators pay national regulators substantial licensing fees to acquire exclusive rights to specific radio spectrum bands. Because the operator owns exclusive access, external devices cannot legally transmit on those frequencies, enabling predictable channel planning, interference minimization, and guaranteed **Quality of Service (QoS)**. End users pay subscription fees (contracts, SIM cards) to access this licensed infrastructure.
    - **Wi-Fi and Bluetooth (Unlicensed):** Operate in free, uncoordinated ISM (Industrial, Scientific, and Medical) bands. Anyone can deploy access points without a license, which results in unpredictable contention and best-effort delivery without carrier-grade QoS guarantees.
- **Service and Session Management:** Because operators guarantee contracted QoS levels and bill for data volume, the cellular core incorporates robust control functions for authentication, dynamic bearer, tunnel establishment, mobility tracking, and policy enforcement.

Network Elements



*In LTE terminology, the radio access component is called the **E-UTRAN** (Evolved Universal Terrestrial Radio Access Network), while the core network is called the **EPC** (Evolved Packet Core).*

1. User Equipment (UE) or Mobile Device: Any terminal device equipped with an LTE radio transceiver (smartphones, tablets, USB modems, connected vehicles, IoT nodes). Each UE contains a SIM (Subscriber Identity Module) card that securely stores the subscriber's credentials, primarily the IMSI (International Mobile Subscriber Identity), a globally unique 64-bit numerical identifier used by cellular networks to identify the subscriber and their home carrier.

2. Base Station (BS): Also named E-NodeB (Evolved Node B) this is located at the physical edge of the carrier's network, managing radio transmissions within its geographical coverage area (the cell). Unlike a standard Wi-Fi Access Point, an eNodeB plays an active, intelligent role in managing radio resources, scheduling, modulation and coordinating device authentication and handovers directly with adjacent base stations.

3. Home Subscriber Server (HSS): The central database residing in the subscriber's home network that stores subscription details, service entitlements, roaming profiles, cryptographic keys, and the subscriber's IMSI. It works closely with the MME to authenticate the UE whenever it attempts to attach to the network.

4. Mobility Management Entity (MME): The primary control-plane signaling node inside the EPC. It does not carry actual user data payloads. Its key functions are**:**

- **Device Authentication:** Validates user credentials with the HSS (mutual device-to-network and network-to-device authentication).
- **Mobility Management:** tracking/paging to locate devices and coordinates the handover between cells/base stations.
- **Tunnel Establishment:** Controls the creation, maintenance, and teardown of data path tunnels between the eNodeB, S-GW, and P-GW.

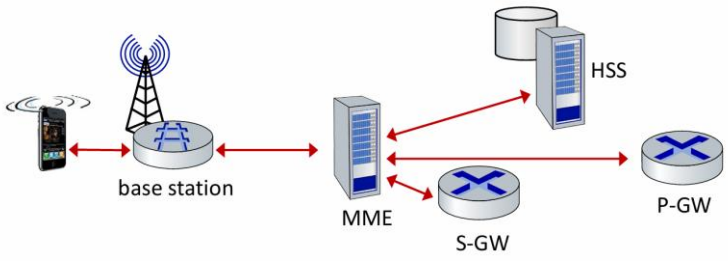
5. Data Gateways (S-GW and P-GW): Together known as the packet gateway architecture, these two elements form the **data-plane** transporting IP packets between the mobile device and external data networks:

- **S-GW (Serving Gateway):** Acts as the local user-plane anchor for data packets. Terminates the user-plane interface from the eNodeB and forwards packets to the P-GW. Anchors the data path during inter-eNodeB handovers, ensuring user data streams do not drop as the UE switches base stations.
- **P-GW (PDN Gateway / Packet Data Network Gateway):** Serves as the boundary router connecting the cellular network to external Packet Data Networks (the public Internet, corporate VPNs, or IMS voice services). Allocates IP addresses to UEs. Enforces QoS policies and provides NAT (Network Address Translation) services. Uses dedicated tunneling protocols to encapsulate and transport user IP packets through the core network.

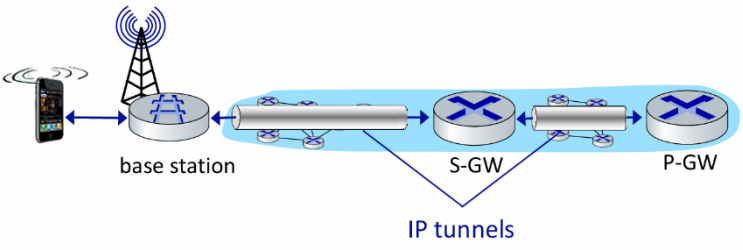
Data Plane vs Control Plane

Cellular architectures (4G LTE and 5G) strictly separate the transmission of signaling instructions from the transmission of user payload traffic. This separation is split into two distinct operational planes:

- **Control Plane:** Manages the administrative and operational state of the connection rather than carrying the user's actual payload. Carries messages for device authentication, connection setup, mobility management, Quality of Service (QoS) negotiation, and tunnel management.



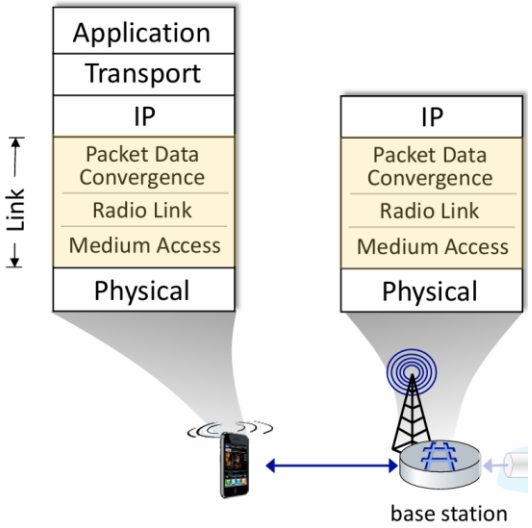
- **Data Plane (or User Plane):** Dedicated exclusively to transporting the user's actual application data (e.g., web pages, streaming video, VoIP packets, file transfers). Focuses on maximizing throughput, minimizing latency, and ensuring packet delivery without inspecting or modifying application payload



LTE Data Plane

As we already said, the LTE Data Plane is responsible for transporting actual application data. To do it, the protocol stack is divided into two main segments:

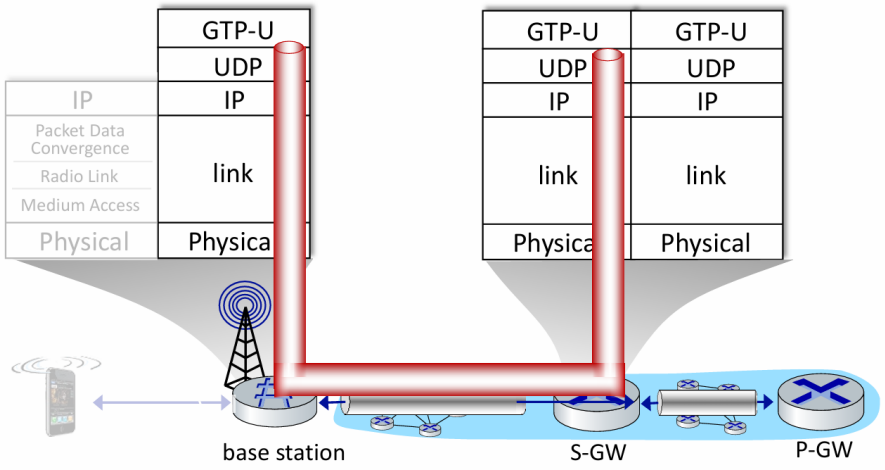
1. **The First Hop:** In wireless cellular networks multiple mobile users share the same radio frequency over an unreliable, noisy channel. To manage this complexity, LTE subdivides Layer 2 into three specialized sub-layers:



- **Packet Data Convergence Protocol (PDCP):** Compresses the upper-layers headers (such as IP, UDP, and TCP) to save valuable radio bandwidth. It also handles the encryption of user data to prevent eavesdropping over the air, as well as integrity protection for signaling messages.
- **Radio Link Control (RLC) Protocol:** Adapts variable-sized IP packets into blocks that fit precisely into the transport block sizes determined by the physical layer. It also performs error detection and selective retransmission to recover from lost or corrupted packets over the radio link.
- **Medium Access Control (MAC) Layer:** Coordinates access to the shared radio spectrum among multiple active users, requesting and allocating radio transmission slots. Provides fast, physical-level error correction and combines logical channels from the RLC layer onto transport channels delivered to the Physical layer.

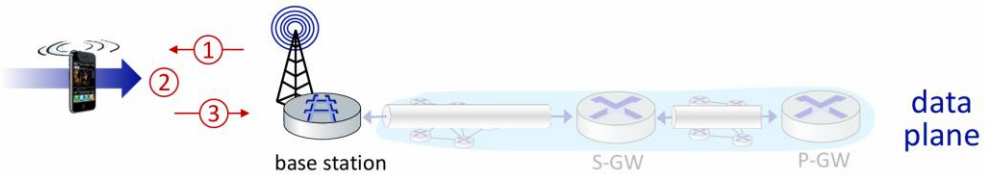
NOTE: On a point-to-point wired link (like ADSL) each subscriber has a dedicated copper line to the Central Office. So, the channel is not shared with anyone. Consequently, ADSL does not require a Medium Access Control (MAC) layer, and its Layer 2 framing is handled simply by protocols like PPP.

2. **The Core Network:** Once the Base Station extracts the user IP datagram from the radio interface, it does not route the packet using standard public IP routing tables. Instead, it utilizes GPT Tunneling:
    a) **The original user IP datagram is encapsulated inside a GTP-U packet**. This GTP packet is further enclosed inside a standard UDP/IP header.
    b) **Then, the Base Station sends the GTP** datagram to the Serving Gateway (S-GW).
    c) **The S-GW re-tunnels the datagram** to the PDN Gateway (P-GW).
    d) **Finally, the P-GW removes the GTP encapsulation**, performs Network Address Translation (NAT) if needed, and forwards the bare user IP datagram onto the public Internet.



NOTE: Mobility via Tunneling. Encapsulating user traffic within GTP tunnels is the foundation of cellular mobility. When a mobile user moves from one Base Station to another, the mobile device retains the exact same public/private IP address assigned by the P-GW. The core network only needs to update the internal tunnel endpoints (switching the destination address of the tunnel from the old BS to the new one). Intermediate core routers and remote web servers on the Internet remain completely unaware that the user has changed physical locations.

Associating with a Base Station



When we are travelling around with our device, it must discover, synchronize with, and select an appropriate base station to establish communication:

1. **Primary Synchronization Signal (PSS) Detection:** Every Base Station continuously broadcasts a PSS every 5 ms on its frequency channel. However, because the mobile device does not know in advance which frequency channel the local cell operates on, the mobile scans through all the frequency bands. Once the device detects a PSS, it receives subframe and slot-level timing synchronization with the transmitter.
2. **Secondary Synchronization Signal (SSS):** On that same detected frequency, the user device (UE) locates the Secondary Synchronization Signal (SSS). Together, the PSS and SSS provide the device with the exact Physical Cell Identity (PCI) and radio frame boundaries. The UE then reads the cell parameters, including the operational channel bandwidth, antenna configuration, and the carrier's PLMN ID (identifying which mobile operator owns the cell).

3. **Cell Selection:** The UE evaluates synchronization signals and system data received from multiple base stations and multiple competing operators. If a shared physical cell site hosts multiple operators, the mobile device filters for its home carrier (or an approved roaming partner). Among available cells of our operator, the mobile selects the base station offering the highest radio signal quality. If the primary cell is congested or degraded, the UE falls back to the next strongest available candidate.
4. **Resource Allocation:** Once the cell is selected, control-plane signaling begins to authenticate the subscriber and establish the GTP tunnels to schedule and allocate specific transmission time-frequency slots.

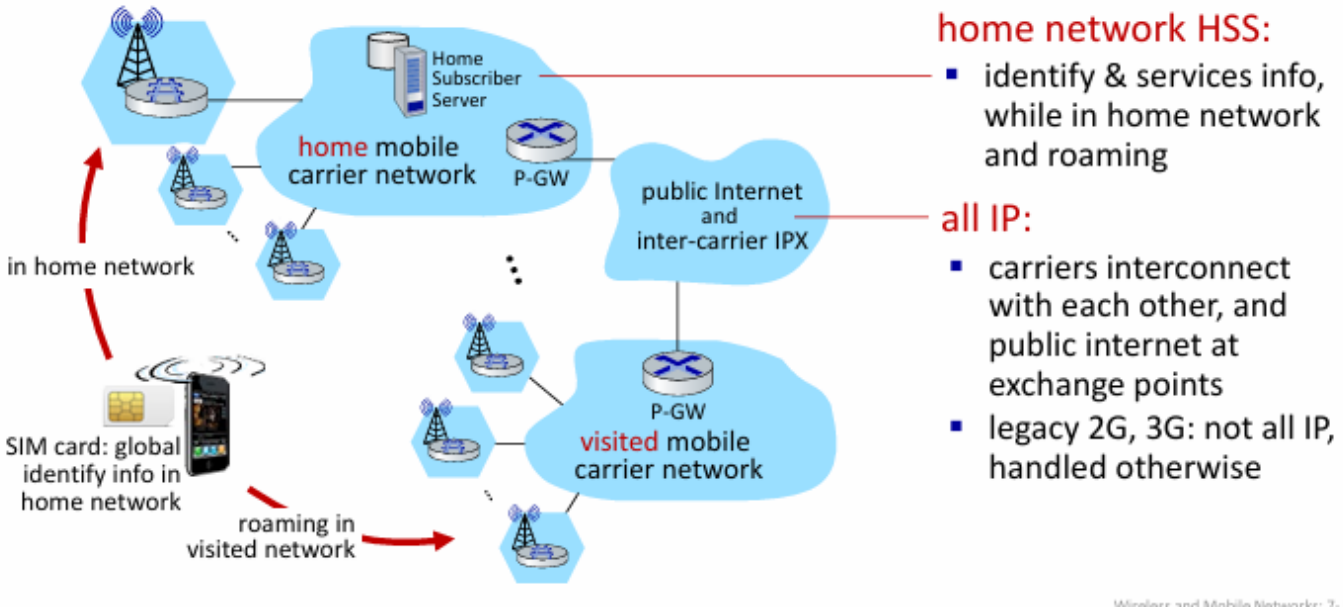
Sleep Modes



The continuous monitoring of the radio channel quickly drains a mobile device's battery. So, to reduce the consumed power, LTE allows mobile phone's cellular modem and radio frequencies to enter a “sleep state” when no data is actively flowing:

- **Light Sleep:** Triggered after a brief period of data inactivity (typically hundreds of milliseconds). The cellular radio sleeps for short intervals, waking up periodically to listen to downstream control channels for incoming data notifications (paging).
- **Deep Sleep:** Triggered after extended periods of inactivity (several seconds). The radio transceiver powers down for longer intervals to maximize battery preservation. Because the user may travel significant distances during deep sleep, the device may move into an entirely different cell coverage area. When the radio wakes up, it need to re-evaluates local radio signals and re-establish the association.

Overall Architecture



Previously, we saw the architecture within a single operator's domain. Now, we want to expands that perspective to show how multiple cellular operators connect to each other. The first main distinction is:

- **Home Mobile Carrier Network:** The network owned by our specific operator. It stores the subscriber's permanent record in the Home Subscriber Server (HSS). The HSS maintains the definitive master profile: user identity, subscribed services, encryption keys, and current location tracking information, both while the subscriber is operating inside the home network and while roaming abroad.
- **Visited Mobile Carrier Network (Roaming):** A network owned and operated by a different carrier. When a user travels outside the geographic coverage of their home network (e.g., traveling abroad or into an area without home carrier infrastructure), the device connects via roaming to a local visited network that maintains a mutual roaming agreement with the home carrier. The visited network temporarily hosts the mobile device, communicating back with the home network's HSS to authenticate the user and authorize data access.

The IP Shift

A fundamental design shift in 4G LTE (and 5G) compared to the previous generations (2G and 3G) is the move to an entirely **All-IP** infrastructure:

- **Legacy Architectures (2G/3G):** Handled voice calls and circuit-switched data through separate, non-IP telephone network elements, while packet data was handled in parallel.
- **Modern Architectures (4G/5G):** Everything, including voice, control signaling, data traffic, etc… is carried natively as IP datagrams.

Mobility Management

Def. Mobility management allows a mobile device to maintain active network sessions while changing its point of attachment to the network

Every subscriber is assigned two primary identifiers:

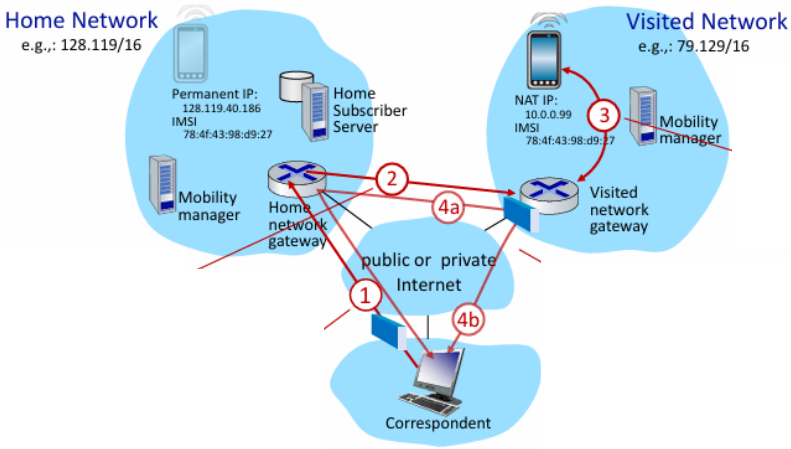
- **International Mobile Subscriber Identity (IMSI):** A permanent, unique hardware-level identity stored inside the SIM card, bound to the user's subscription profile in the Home Subscriber Server (HSS).
- **Permanent Home IP Address:** A static or persistent IP address assigned by the mobile device's home network operator. External correspondents send packets to this permanent address.

When a mobile device travels outside the geographic reach of its home carrier and enters a **visited network**:

1. **Association:** The device associates with a local base station and identifies the visited mobility manager.
2. **Registration with Home HSS:** IF there is a roaming agreement between the two operators, the visited mobility manager contacts the subscriber's home HSS.
3. **Authentication:** The home HSS authenticates the subscriber's SIM credentials, authorizes service access, and records the mobile device's current location (the address of the visited network gateway).

Once registered in the visited network, there are two primary routing strategies to deliver packets between an external correspondent and the roaming mobile device: **Indirect Routing** and **Direct Routing**.

Indirect Routing



In indirect routing (also named Triangle Routing), the correspondent remains completely unaware that the mobile user has moved to a visited network. All packets are routed via the user's home network:

(1) **The correspondent addresses an IP datagram to the permanent home IP address**. Standard Internet routing delivers this datagram to the mobile user's home gateway.

(2) **The home gateway intercepts the datagram**, looks up the mobile's current location registered with the HSS, encapsulates the packet inside a tunnel and forwards it to the visited network gateway.

(3) **The visited gateway router decapsulates the tunnel** and delivers the original datagram over the radio link to the mobile user.

(4) When the mobile user replies, traffic can follow one of two paths:

a. **Reverse Tunneling:** The visited gateway forwards the reply back to the home gateway, which forwards it to the correspondent.

b. **Direct Reply:** The mobile sends the packet directly to the correspondent's IP address through the visited network router.

Advantages:

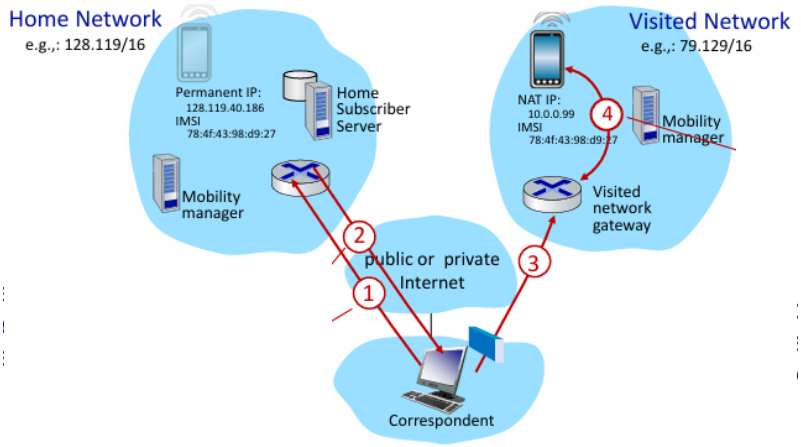
- **Completely Transparent:** The correspondent requires no special software or awareness of the user's physical location; it simply sends traffic to the home IP address.
- **Session Continuity:** Ongoing transport-layer connections (e.g., TCP) remain intact while the mobile user moves across different visited networks, because the user's visible IP address never changes.

Disadvantages:

- Datagrams follow an indirect triangular path, introducing unnecessary propagation delay.

Example: Suppose your home network operator is in Rome, and you travel to Berlin to visit your girlfriend, who uses a local German network. If she sends you a message while you are sitting in the same room in Berlin, her message must first travel across Europe to your home gateway in Rome **(1)**, which tunnels it all the way back to Berlin **(2)**, before reaching your phone **(3)**. This routing detour is highly inefficient…

Direct Routing



Direct routing eliminates the triangular detour by allowing the correspondent to send packets straight to the visited network:

(1)(2) Before transmitting, the correspondent contacts the mobile's home HSS, providing the mobile's identity and requesting its current visited network address. The home HSS returns the visited network gateway's address.

(3) The correspondent addresses and transmits its datagrams directly to the visited network address.

(4) The visited gateway router receives the incoming datagram and delivers it to the mobile user.

Advantages:

- **Optimal Path:** Overcomes triangle routing inefficiencies by transferring data directly between endpoints over the shortest available path.

Disadvantages:

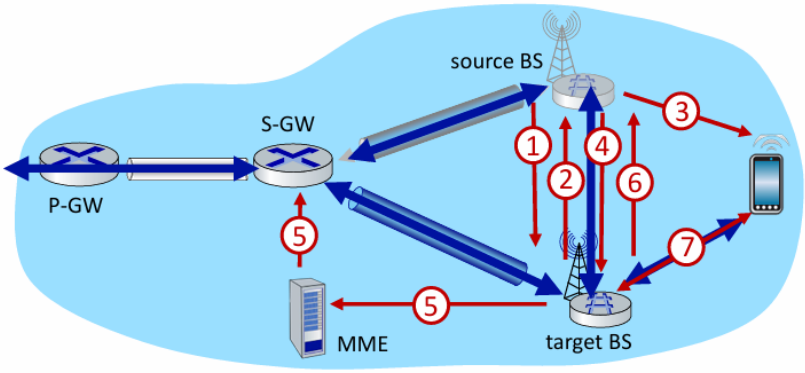
- **Non-Transparent to Correspondent:** The correspondent host or edge router must support specialized mobility protocols to query the home registry before initiating communication.
- **Privacy Exposure:** The home network must expose the mobile subscriber's current geographic location and network point of attachment to third parties.
- **Handoff Complexity During Active Sessions:** If the mobile user changes visited networks while a communication session is active, additional forwarding mechanisms must be established between the old visited network and the new visited network to avoid dropping the connection.

Handover / Handoff

While the mobility management manages the routing traffic between different operators, the handover (or handoff) mechanism operates at the radio access layer.

Its goal is to dynamically migrate an active user session from one base station to another as the subscriber moves through the network. In an active handover scenario, two base stations are involved:

- **Source Base Station:** The current base station serving the User Equipment (UE).
- **Target Base Station:** The new base station to which the session will be transferred.



In 4G LTE, handovers are network-controlled and mobile-assisted, meaning that the mobile device itself continuously measures the radio signal strength of the neighboring cells, reporting these metrics back to the source base station. When signal conditions satisfy handover criteria, the handover procedure starts:

1. **Handover Request:** Based on the user's measurement reports, the source BS selects the optimal target base station. Then, the source BS sends a Handover Request to the target BS. This message includes the UE's current context, active configurations, and Quality of Service requirements.
2. **Admission:** The target BS evaluates whether it has sufficient radio resources and capacity to accept the incoming user. If accepted, the target BS pre-allocates radio time-frequency slots and responds with a **Handover Request Acknowledgment (HR ACK)**, including the access parameters needed for the UE to attach to the new cell.
3. **Handover Command:** The source BS forwards the configuration to the mobile device in a Handover Command. At this stage, the mobile switches its radio transceiver to the target operating frequency and begins transmitting to the new base station. From the mobile user's perspective, the handover is now functionally complete.
4. **Packet Forwarding:** To prevent data loss during the switch phase, the source BS stops transmitting downlink datagrams to the mobile device. Instead, the source BS buffers incoming packets and immediately forwards (tunnels) them to the target BS, which delivers them to the mobile device over the new radio channel.
5. **Path Switch:** The target BS notifies the **MME** that it is now the active base station for the mobile subscriber, though a **Path Switch Request**. The MME instructs the **Serving Gateway (S-GW)** to update its user-plane tunnel endpoint.
6. **Resource Release:** Once the core network path switch is acknowledged, the target BS sends a UE Context Release message back to the source BS, so that it can releases the radio and memory resources previously allocated to that mobile user.
7. **Conclusion:** At this point, the user datagrams flow directly through the new GTP tunnel between the target node and the S-GW, without passing through the source base station.

Alternative: modern deployments can use coordinated mechanisms where a cluster of base stations cooperates to track the mobile device's position and channel state to proactively choose the ideal target cell, optimizing overall network load and interference.

4G vs 5G

Compared to 4G LTE, 5G is engineered around three target performance benchmarks:

- **10× Increase in Peak Bitrate:** Delivering up to 10-20 GBps peak downstream throughput.
- **10× Decrease in Latency:** Reducing round-trip radio latency to roughly 1 ms (enabling mission-critical, real-time communications).
- **100× Increase in Traffic Capacity:** Supporting massive device densities to accommodate the Internet of Things (IoT).

To do it, 5G introduces a completely new air interface called **5G NR (New Radio)**, which is not backwards-compatible with 4G LTE and introduces severe path loss, atmospheric attenuation, and zero penetration through solid obstacles. To compensate these problems, the 5G base stations utilize **Massive MIMO** (Multiple-Input Multiple-Output) arrays with dozens or hundreds of directional antenna elements.

Instead of broadcasting RF power across an entire cell sector, the base station focuses narrow, highly concentrated beams of electromagnetic energy directly toward individual mobile devices. However, it requires thousands of small base stations (**pico-cells** and **micro-cells**) across street lamps, utility poles, and indoor venues.

In addition, while 4G relies on a centralized core hardware, 5G distributes its core functions into a **distributed software**. User-plane processing and authentication are pushed to the physical network edge (Multi-access Edge Computing - MEC) close to base stations, drastically cutting latency.